Assignment 1: Mastering Azure Data Factory (ADF) End-to-End

Assignment Objective

By completing this assignment, students must:

- Understand Azure Data Factory architecture
- Build real-world pipelines
- Implement dynamic and advanced workflows
- Deploy and monitor solutions
- Produce professional-level technical documentation

Important:
By the end of this assignment, your documentation must be so clear and complete that a new student can master ADF without watching any video.

Learning Resources (Mandatory)

Phase 1 & 2 (Beginner → Intermediate)

Follow this video step by step:

Azure Data Factory Full Course (Beginner to Pro)
https://www.youtube.com/watch?v=8zIVOdKyoDA

Phase 3 (Advanced Concepts)

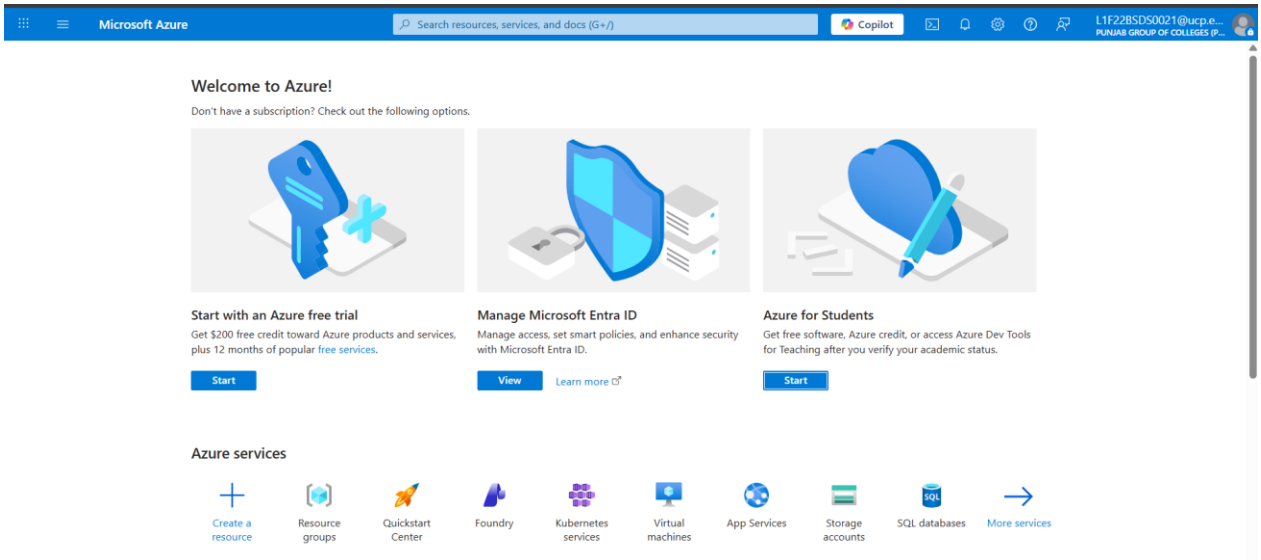
Follow this advanced guide:

Azure Data Factory Advanced Course – Complete Guide
https://www.youtube.com/watch?v=KxUm1eXQha8

Part 1 – Fundamentals (Based on Video 1)

1. Azure Setup & ADF Creation

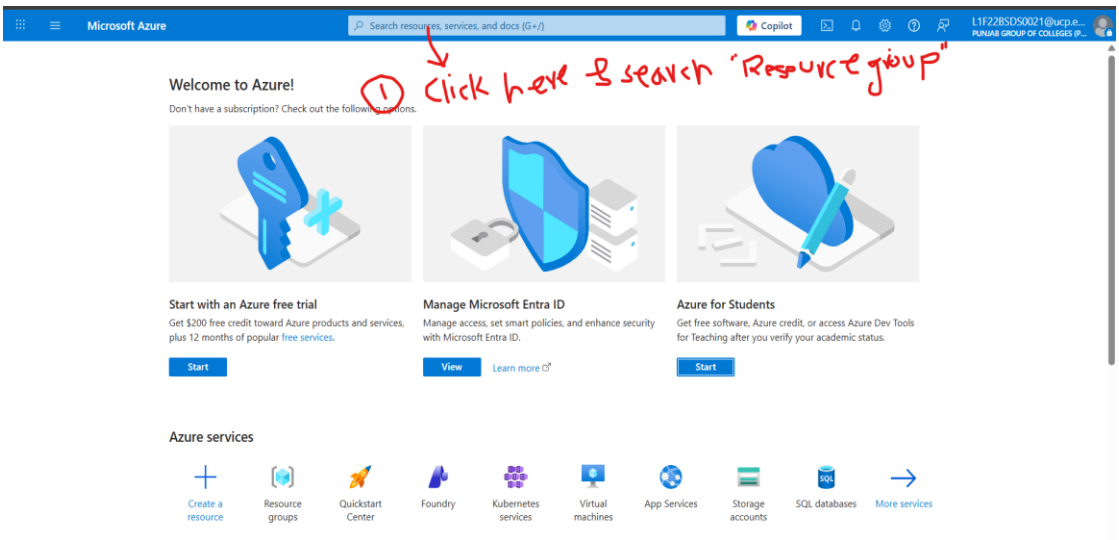
- Create Azure account



- Create Resource Group
- Create Azure Data Factory
- Explore Author / Monitor / Manage tabs

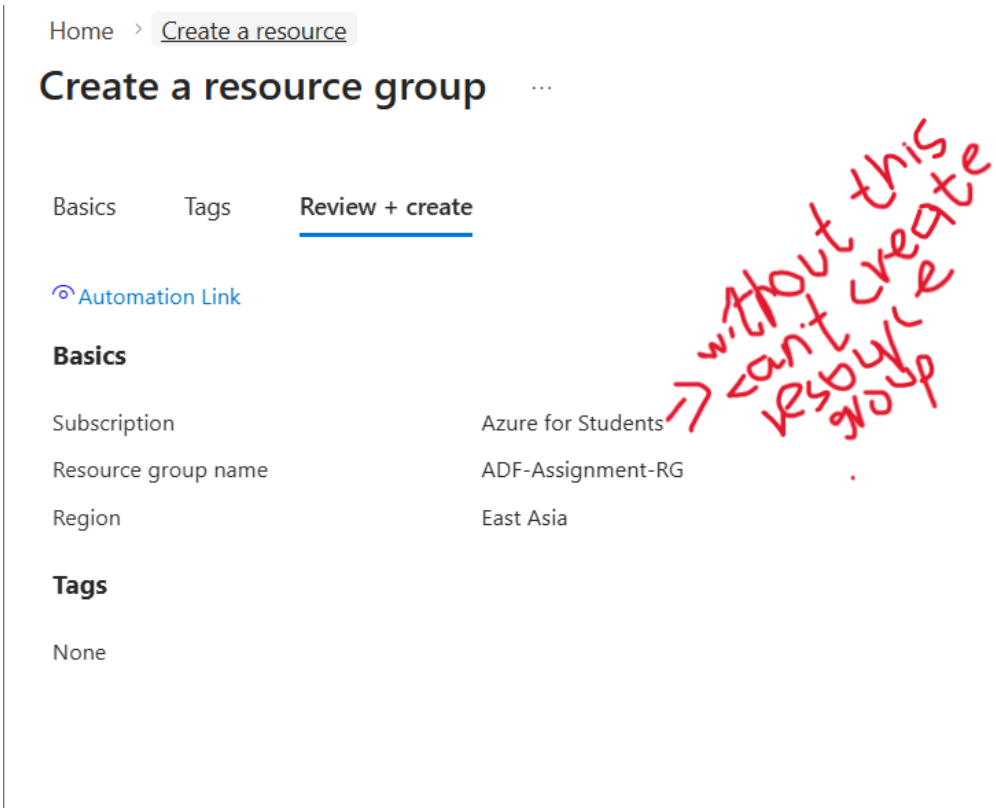
Deliverables:

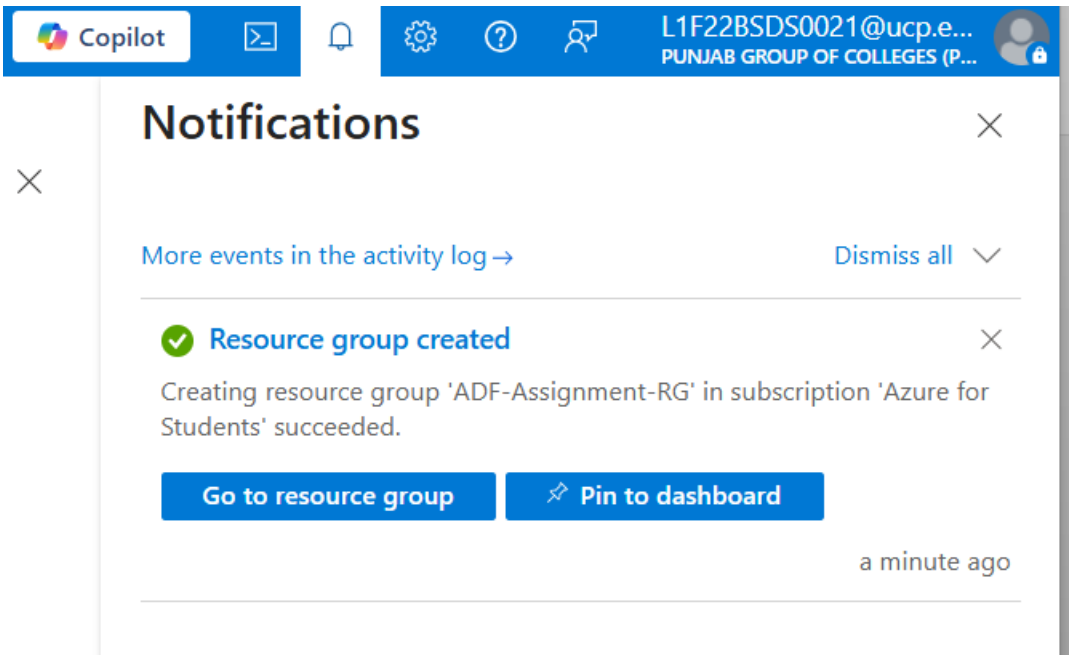
- Screenshots of:
  - Resource Group

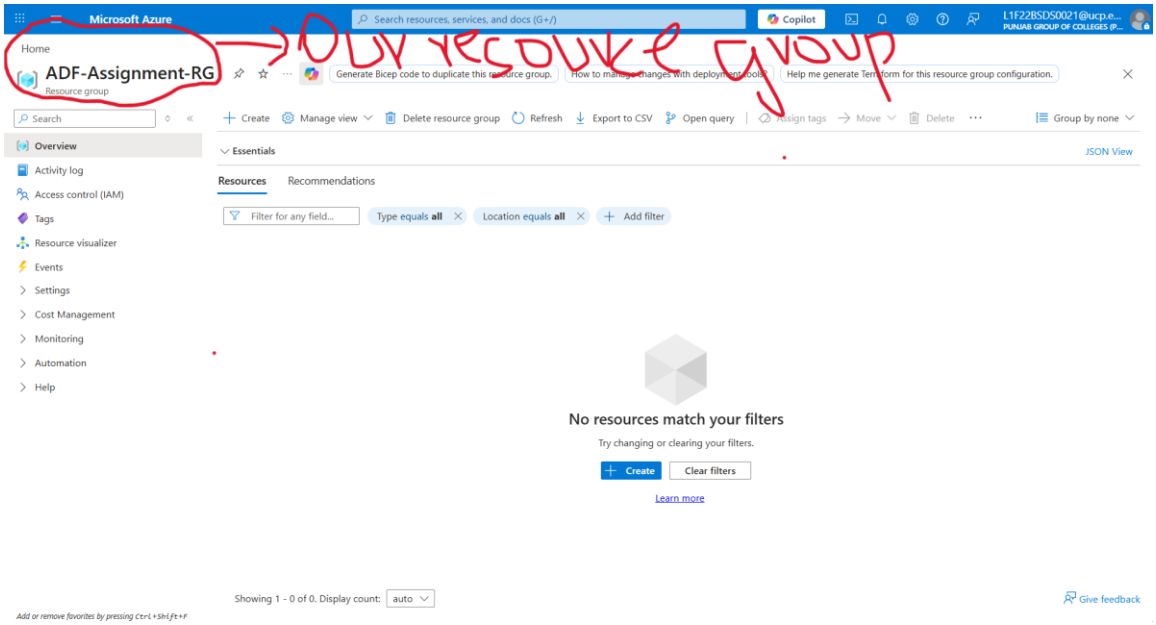


2) Then,Click resource group one with “folder” shape icon,

3) Next, we have to specify the subscription we want to use for this purpose and then name our resource group.







Before we move to ADF we a place where to store files. We are creating storage azure blob

As follows:

Home

Create a storage account

Project details

Select the subscription in which to create the new storage account. Choose a new or existing resource group to organize and manage your storage account together with other resources.

Subscription * Azure for Students

Resource group * ADF-Assignment-RG

Create new

storage acc

Instance details

Storage account name * adfstorageahmad

Region * (Asia Pacific) East Asia

Deploy to an Azure Extended Zone

Validation could not be completed due to an error. If this issue persists, please contact support.

Preferred storage type Choose preferred storage type

This helps us provide relevant guidance. It doesn't restrict your storage to this resource type. Learn more

Select LRS

Performance *

- **Standard:** Recommended for most scenarios (general-purpose v2 account)
- **Premium:** Recommended for scenarios that require low latency.

Redundancy * Locally-redundant storage (LRS)

Previous | Next | Review + create

Give feedback

Home

adfstorageahmad_1774635661298 | Overview

Deployment

Search

- Overview
- Inputs
- Outputs
- Template

Delete | Cancel | Redeploy | Download | Refresh

Deployment is in progress

Deployment name: adfstorageahmad_1774635661298
Subscription: Azure for Students
Resource group: ADF-Assignment-RG

Start time: 3/27/2026, 11:21:47 PM
Correlation ID: c86c895c-0b52-499b-a0c3-41765e3e3c2a

Deployment details

Resource	Type	Status	Operation details
adfstorageahmad	Microsoft.Storage/storageAccou...	Accepted	Operation details

Give feedback

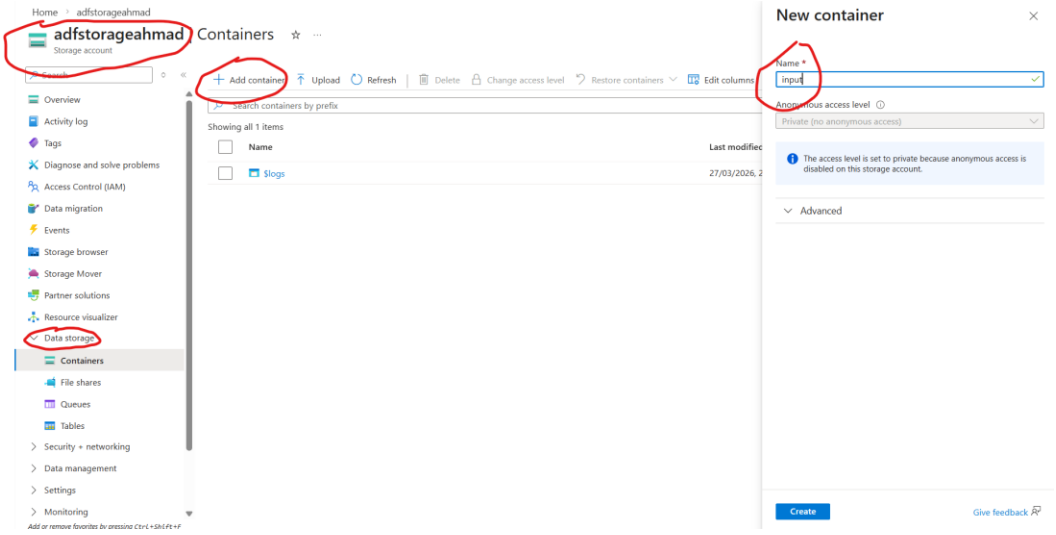
Tell us about your experience with deployment

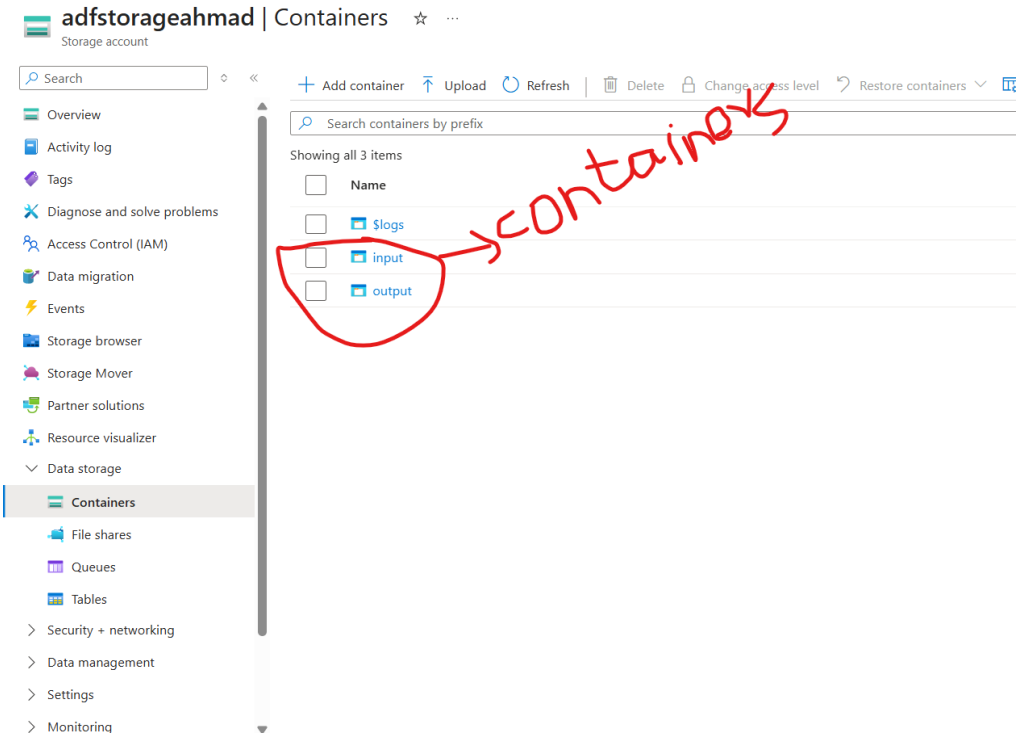
Microsoft Defender for Cloud
Secure your apps and infrastructure
Go to Microsoft Defender for Cloud >

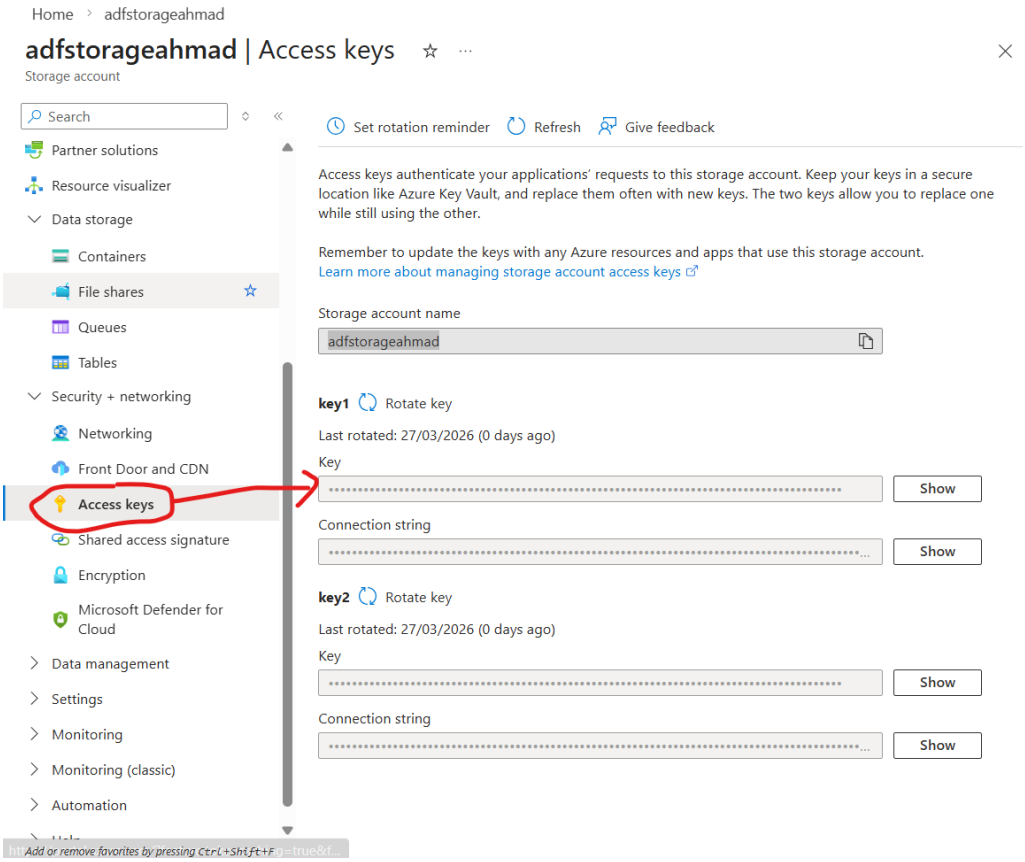
Free Microsoft tutorials
Start learning today >

Work with an expert
Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support.
Find an Azure expert >

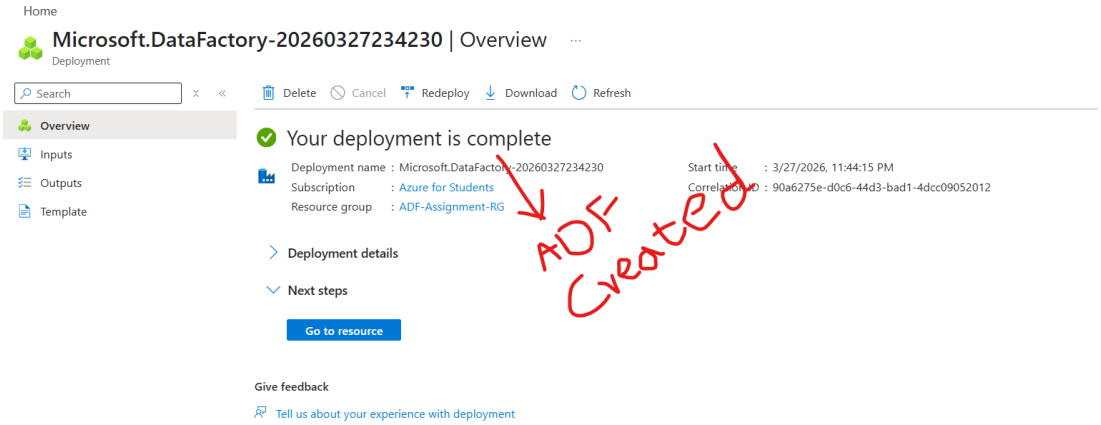
Input Container



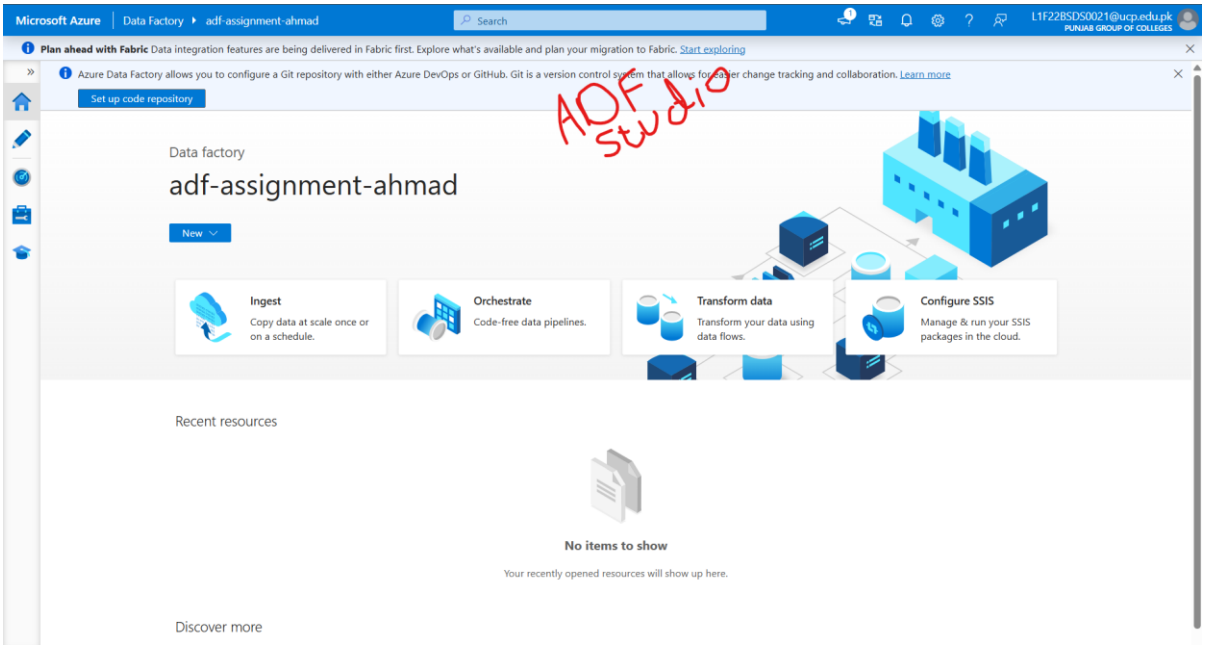



Home > adfstorageahmad
adfstorageahmad | Access keys
Storage account
Search
Partner solutions
Resource visualizer
Data storage
Containers
File shares
Queues
Tables
Security + networking
Networking
Front Door and CDN
Access keys
Shared access signature
Encryption
Microsoft Defender for Cloud
Data management
Settings
Monitoring
Monitoring (classic)
Automation
Set rotation reminder
Refresh
Give feedback
Access keys authenticate your applications' requests to this storage account. Keep your keys in a secure location like Azure Key Vault, and replace them often with new keys. The two keys allow you to replace one while still using the other.
Remember to update the keys with any Azure resources and apps that use this storage account.
Learn more about managing storage account access keys
Storage account name
adfstorageahmad
key1
Rotate key
Last rotated: 27/03/2026 (0 days ago)
Key
Show
Connection string
Show
key2
Rotate key
Last rotated: 27/03/2026 (0 days ago)
Key
Show
Connection string
Show

ADF CREATED:


Home
Microsoft.DataFactory-20260327234230 | Overview
Deployment
Search
Overview
Inputs
Outputs
Template
Delete
Cancel
Redeploy
Download
Refresh
Your deployment is complete
Deployment name : Microsoft.DataFactory-20260327234230
Subscription : Azure for Students
Resource group : ADF-Assignment-RG
Start time : 3/27/2026, 11:44:15 PM
Correlation ID : 90a6275e-d0c6-44d3-bad1-4dcc09052012
ADF Created
Deployment details
Next steps
Go to resource
Give feedback
Tell us about your experience with deployment

- ADF Studio Interface



- Explanation in your Word document:

What is ADF?

Answer: Azure Data Factory (ADF) is a cloud-based data integration service provided by Microsoft that allows you to collect, transform, and move data from different sources to destinations.
It helps in creating data pipelines to automate data workflows.

Why is it used?

ADF is used for:

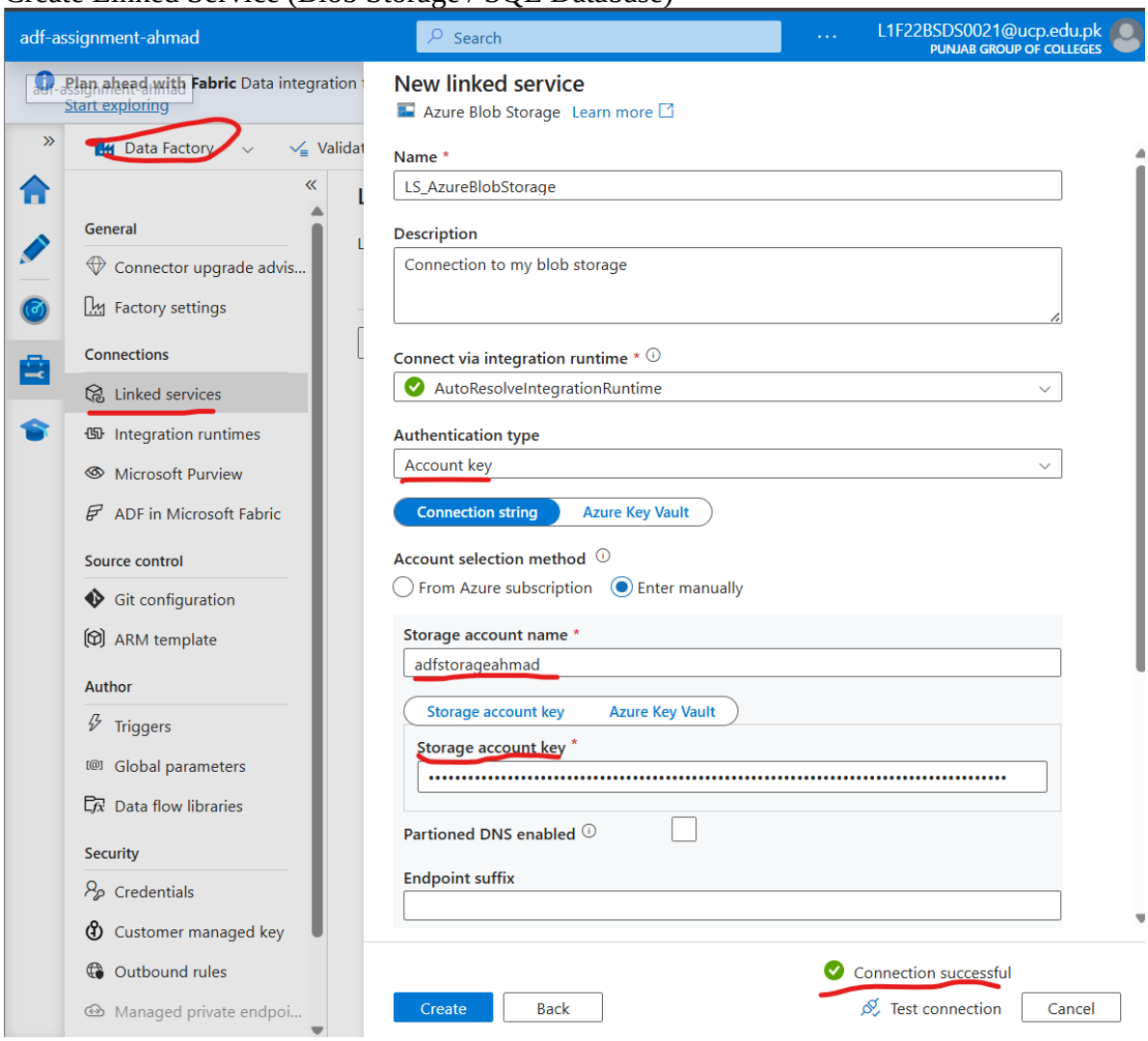
- **Data Integration:** Combine data from multiple sources (databases, APIs, files)
- **Data Movement:** Copy data from one place to another (e.g., Blob → SQL)
- **Data Transformation:** Clean and process data using Data Flows or external tools
- **Automation:** Run pipelines automatically using triggers (schedule/event-based)
- **Cloud-based Processing**: No need for local infrastructure

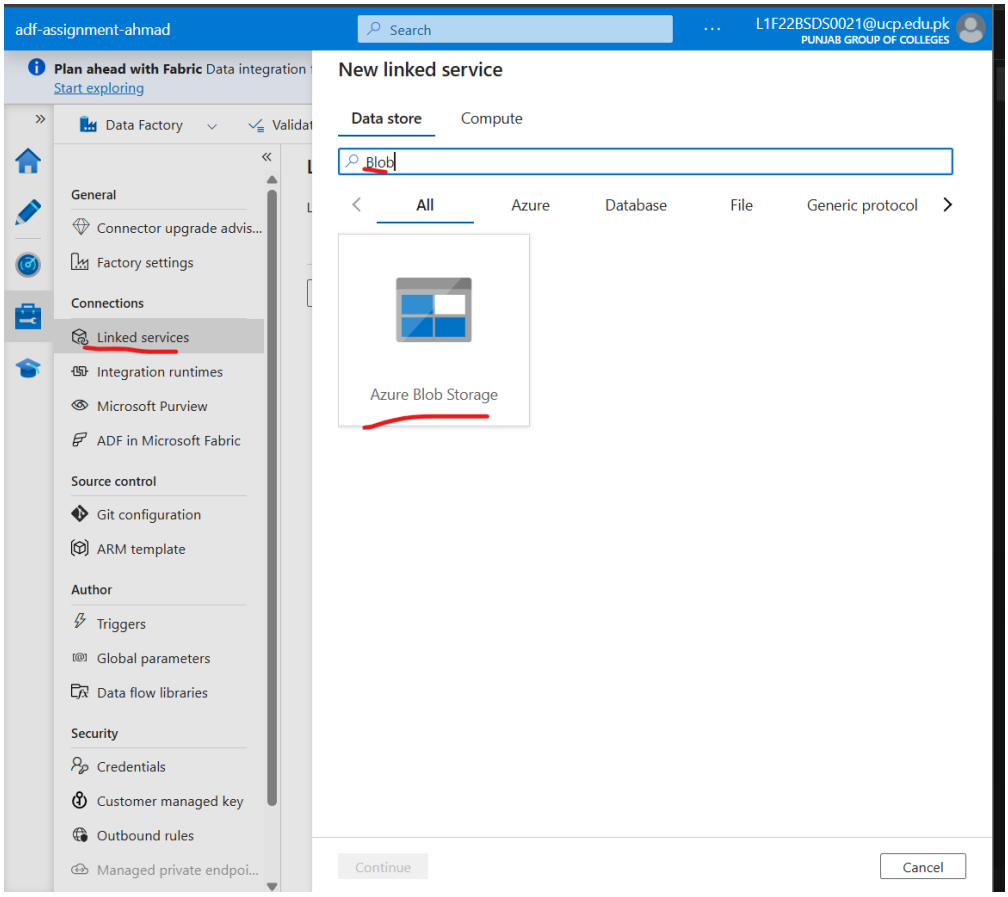
ADF is used to automate data handling and processing in the cloud.

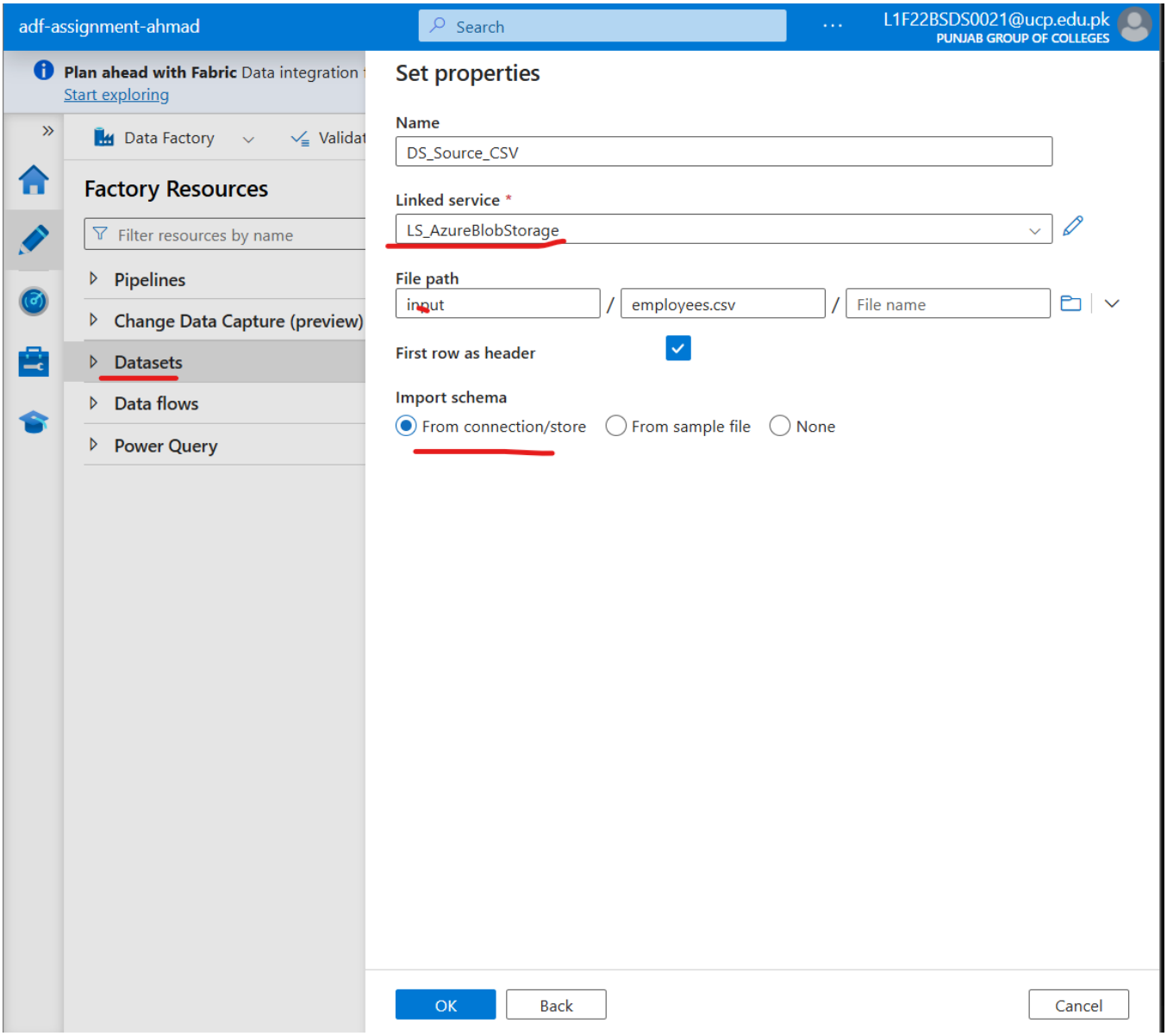
Difference between ETL and ELT

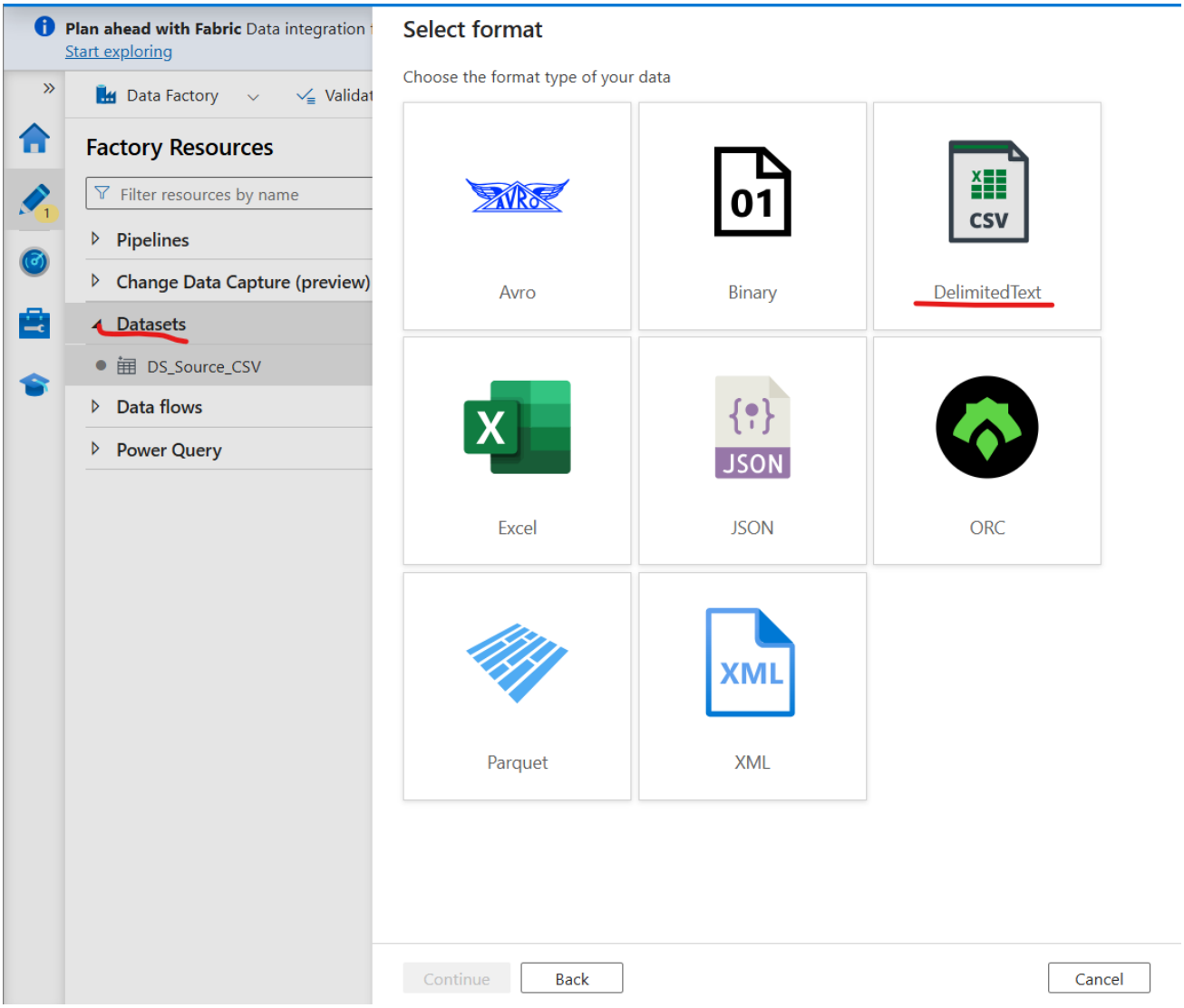
Feature	ETL (Extract, Transform, Load)	ELT (Extract, Load, Transform)
Order	Transform before loading	Transform after loading
Process	Data is cleaned first	Raw data is loaded first
Speed	Slower (extra step before load)	Faster (load first, then process)
Storage	Needs staging area	Uses data warehouse power
Example	Traditional systems	Modern tools like ADF, Snowflake

2. Linked Services & Datasets

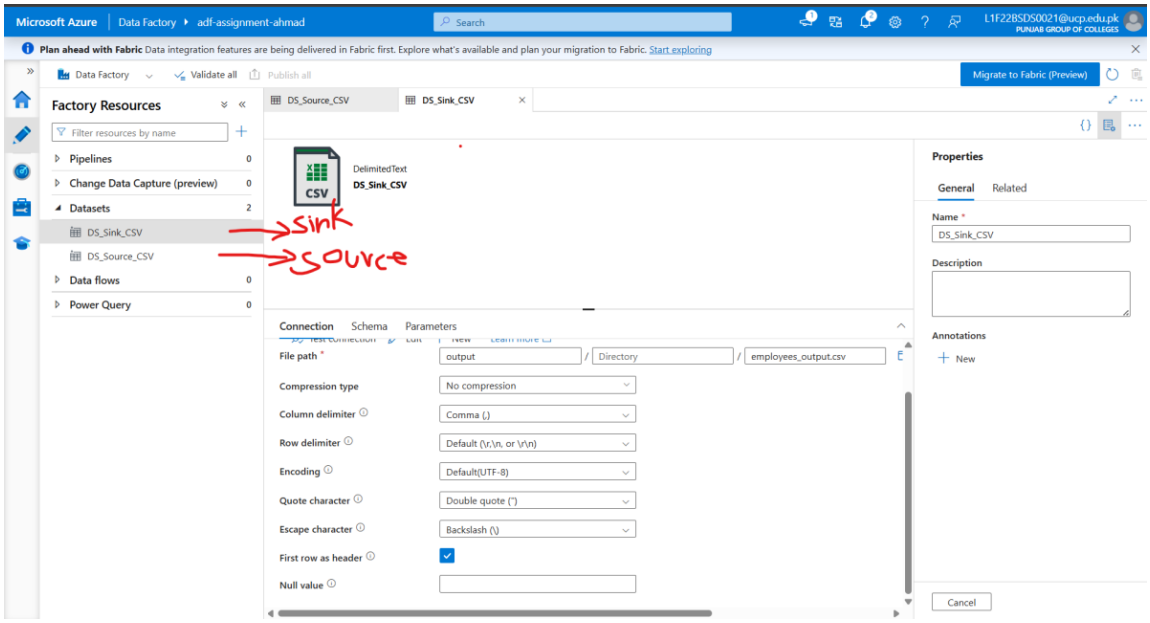
- Create Linked Service (Blob Storage / SQL Database)
- 

- 
adf-assignment-ahmad
Search
L1F22BSDS0021@ucp.edu.pk
PUNJAB GROUP OF COLLEGES
Plan ahead with Fabric Data integration
Start exploring
Data Factory
General
Connector upgrade advis...
Factory settings
Connections
Linked services
Integration runtimes
Microsoft Purview
ADF in Microsoft Fabric
Source control
Git configuration
ARM template
Author
Triggers
Global parameters
Data flow libraries
Security
Credentials
Customer managed key
Outbound rules
Managed private endpoi...
New linked service
Data store
Compute
Blob
All
Azure
Database
File
Generic protocol
Azure Blob Storage
Continue
Cancel

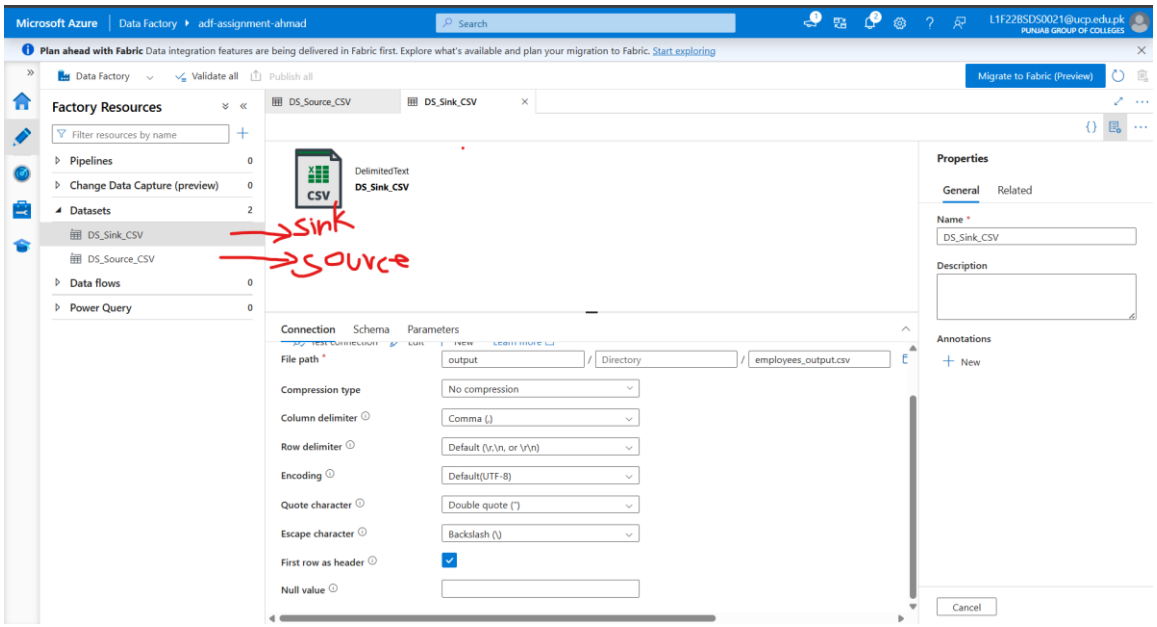

adf-assignment-ahmad
Search
L1F22BSDS0021@ucp.edu.pk
PUNJAB GROUP OF COLLEGES
Plan ahead with Fabric Data integration
Start exploring
Data Factory
Factory Resources
Filter resources by name
Pipelines
Change Data Capture (preview)
Datasets
Data flows
Power Query
Set properties
Name
DS_Source_CSV
Linked service *
LS_AzureBlobStorage
File path
input
employees.csv
File name
First row as header
Import schema
From connection/store
From sample file
None
OK
Back
Cancel



- Create Source Dataset

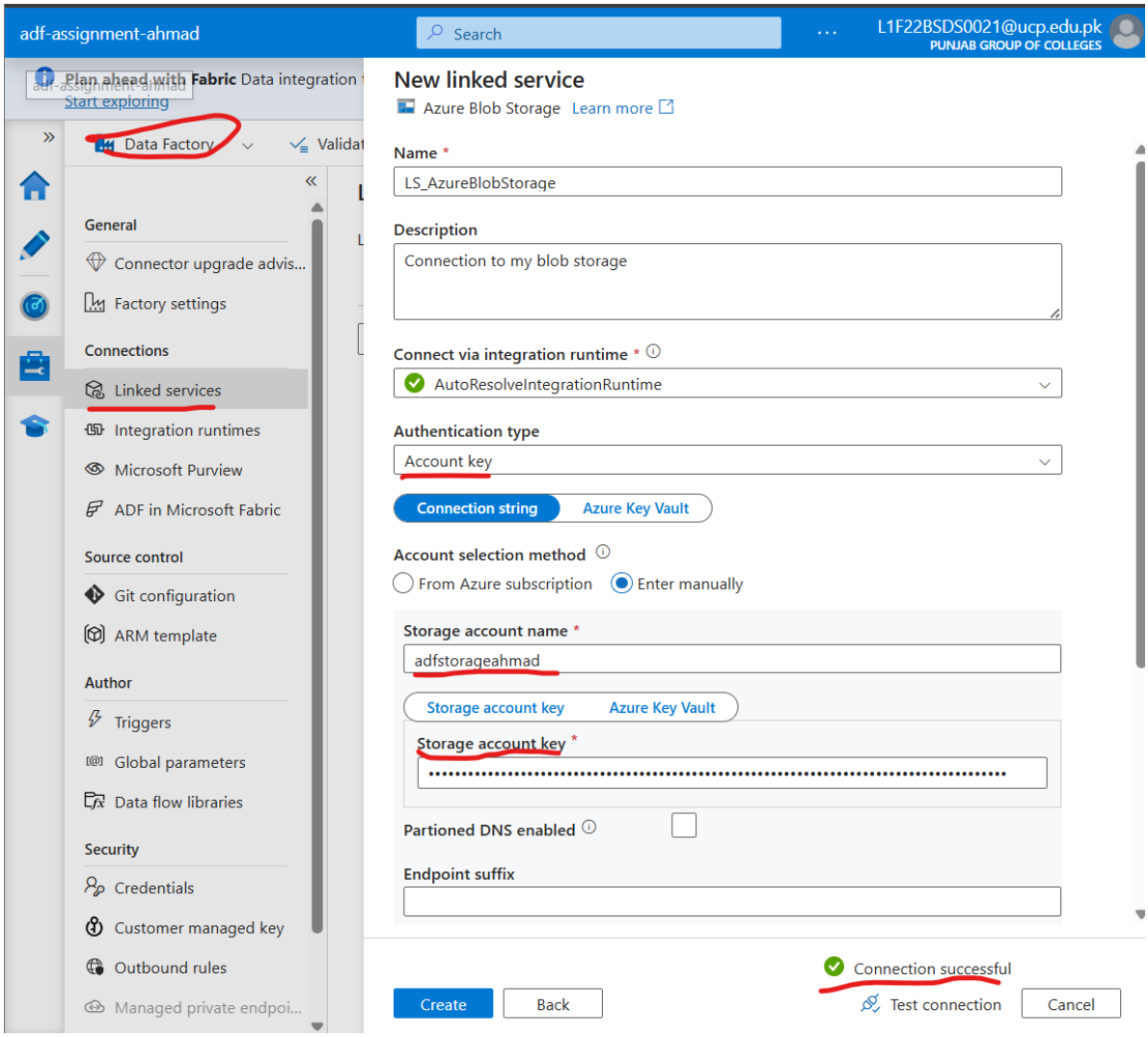


- Create Sink Dataset



Deliverables:

- Screenshots of Linked Services configuration



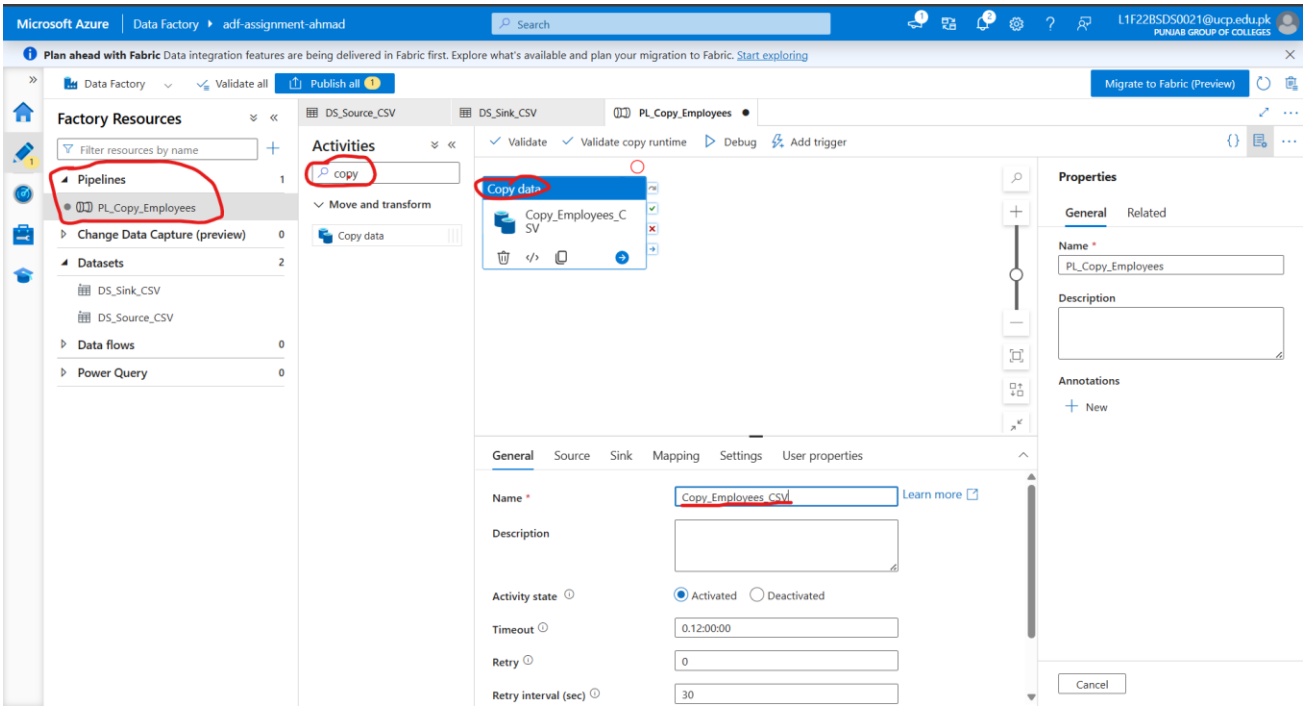
- Explanation of:
  - What is a Linked Service?
  - What is a Dataset?
  - Relationship between them

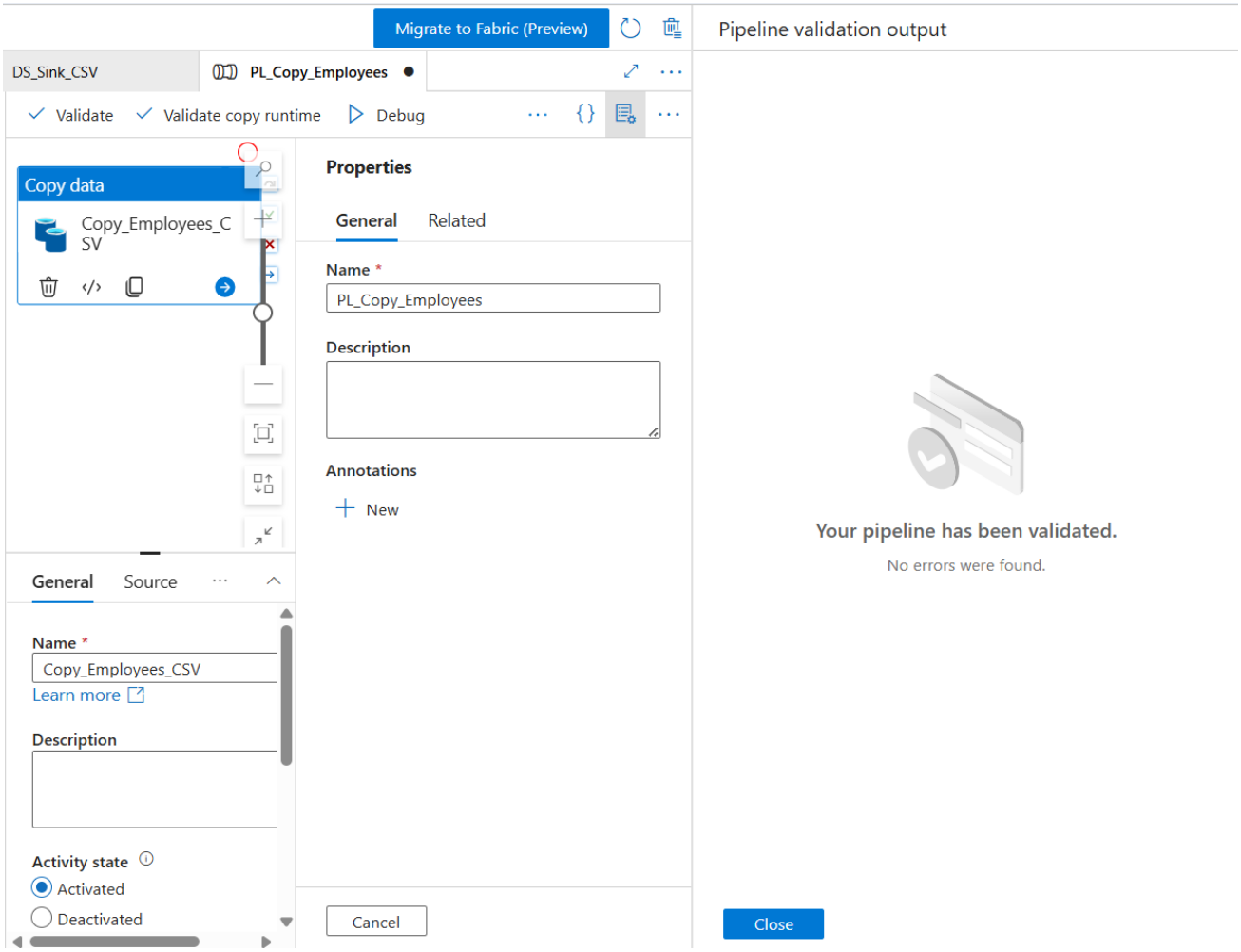
3. First Pipeline – Copy Activity

- Create Copy Data pipeline
- Run in Debug mode
- Trigger pipeline manually

Deliverables:

- Pipeline design screenshot
- Run history screenshot
- Write detailed explanation:
  - Step-by-step creation
  - Common errors and how you solved them
  - What happens internally during Copy Activity


Microsoft Azure
Data Factory
adf-assignment-ahmad
Search
L1F22BSDS0021@ucp.edu.pk
PUNJAB GROUP OF COLLEGES
Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. Start exploring
Data Factory
Validate all
Publish all 1
Migrate to Fabric (Preview)
Factory Resources
Filter resources by name
Pipelines
PL_Copy_Employees
Change Data Capture (preview)
Datasets
DS_Sink_CSV
DS_Source_CSV
Data flows
Power Query
DS_Source_CSV
DS_Sink_CSV
PL_Copy_Employees
Activities
copy
Move and transform
Copy data
Validate
Validate copy runtime
Debug
Add trigger
Copy data
Copy_Employees_CSV
Properties
General
Related
Name *
PL_Copy_Employees
Description
Annotations
New
General
Source
Sink
Mapping
Settings
User properties
Name *
Copy_Employees_CSV
Learn more
Description
Activity state
Activated
Deactivated
Timeout
0.12:00:00
Retry
0
Retry interval (sec)
30
Cancel


Migrate to Fabric (Preview)
Pipeline validation output
DS_Sink_CSV
PL_Copy_Employees
Validate
Validate copy runtime
Debug
Copy data
Copy_Employees_CSV
Properties
General
Related
Name *
PL_Copy_Employees
Description
Annotations
New
General
Source
Name *
Copy_Employees_CSV
Learn more
Description
Activity state
Activated
Deactivated
Cancel
Your pipeline has been validated.
No errors were found.
Close

Output ...

Pipeline run ID e71c03ee-7e09-4499-9 7a2-cdf95b28a334

Pipeline status Succeeded

View debug run consumption

Monitor in Azure Metrics

Export to CSV

Showing 1 - 1 of 1 items

Activity name
Copy_Employees_CSV

Details Refresh

Copy Pipeline

Learn more on copy performance details from here.

Activity run id: e19605dc-df56-4ceb-a9f8-c0a552212a84

Azure Blob Storage
Region: Central India

Succeeded

Azure Blob Storage
Region: Central India

Data read:	96 bytes	**Data written:**	96 bytes
Files read:	1	**Files written:**	1
Peak connections:	1	**Peak connections:**	1

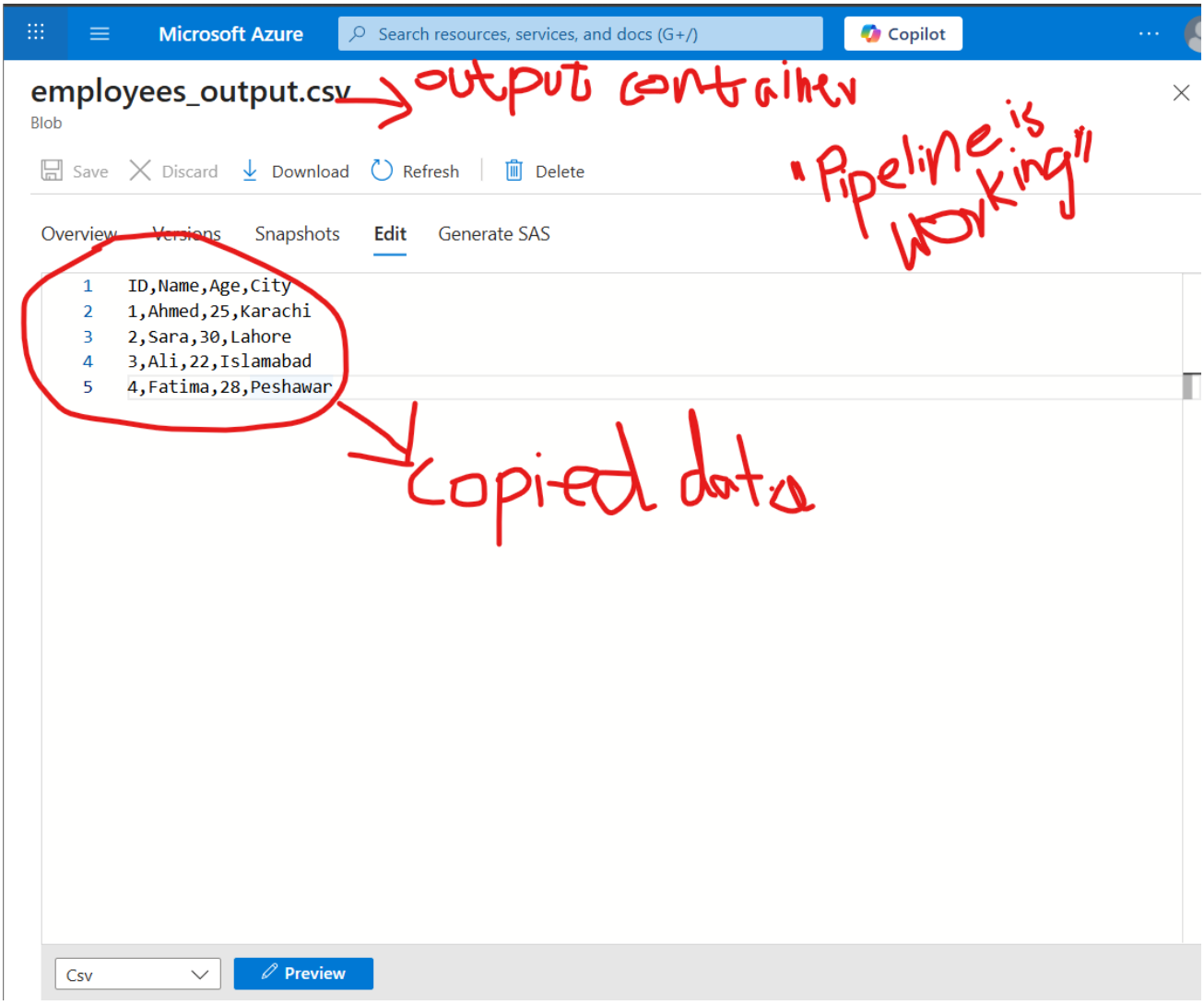
Copy duration	00:00:10
Throughput:	48 bytes/s

Azure Blob Storage → Azure Blob Storage

Start time	3/28/2026, 9:19:54 AM
Used DIUs	4
Used parallel copies	1
Duration	00:00:10

Details		Working duration	Total duration
Queue			00:00:06
Transfer	Listing source	00:00:00	00:00:02
	Reading from source	00:00:00	
	Writing to sink	00:00:00	

Data consistency verification Not verified

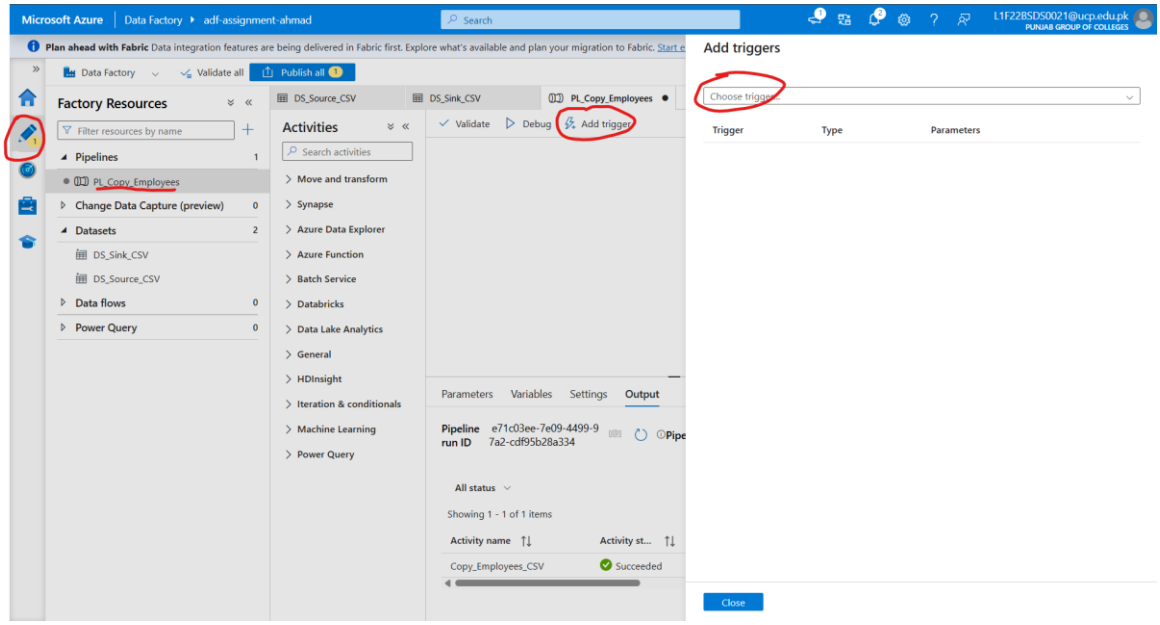


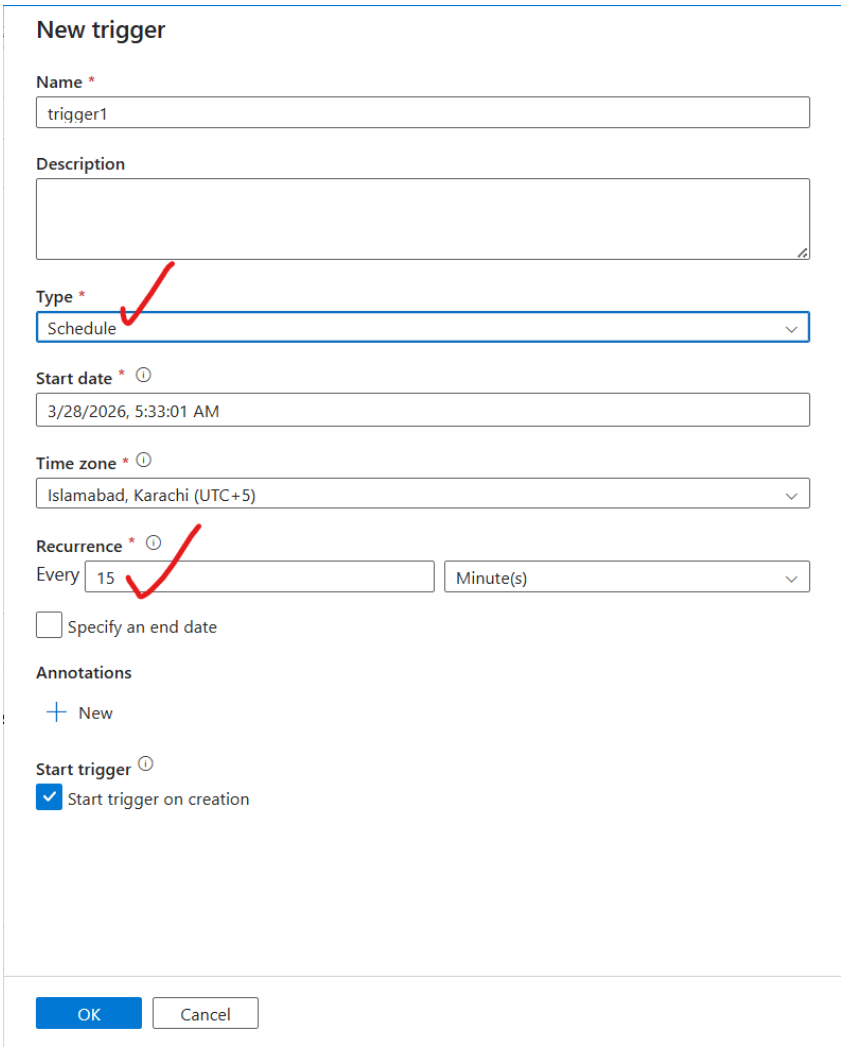
4. Triggers

- Create Schedule Trigger
- Understand manual vs scheduled trigger

Deliverables:

- Trigger configuration screenshot
- Explanation:
  - Difference between manual, schedule, and event trigger


Microsoft Azure
Data Factory
adf-assignment-ahmad
Search
L1F22BSDS0021@ucp.edu.pk
PUNJAB GROUP OF COLLEGES
Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric.
Data Factory
Validate all
Publish all
Factory Resources
Filter resources by name
Pipelines
PL_Copy_Employees
Change Data Capture (preview)
Datasets
DS_Sink_CSV
DS_Source_CSV
Data flows
Power Query
DS_Source_CSV
DS_Sink_CSV
PL_Copy_Employees
Activities
Search activities
Move and transform
Synapse
Azure Data Explorer
Azure Function
Batch Service
Databricks
Data Lake Analytics
General
HDInsight
Iteration & conditionals
Machine Learning
Power Query
Validate
Debug
Add trigger
Parameters
Variables
Settings
Output
Pipeline run ID
e71c03ee-7e09-4499-9 7a2-cdf95b28a334
All status
Showing 1 - 1 of 1 items
Activity name
Activity st...
Copy_Employees_CSV
Succeeded
Add triggers
Choose trigger
Trigger
Type
Parameters
Close


New trigger
Name *
trigger1
Description
Type *
Schedule
Start date *
3/28/2026, 5:33:01 AM
Time zone *
Islamabad, Karachi (UTC+5)
Recurrence *
Every
15
Minute(s)
Specify an end date
Annotations
New
Start trigger
Start trigger on creation
OK
Cancel

Part 2 – Intermediate Concepts (Video 1 Continued)

5. Control Flow Activities

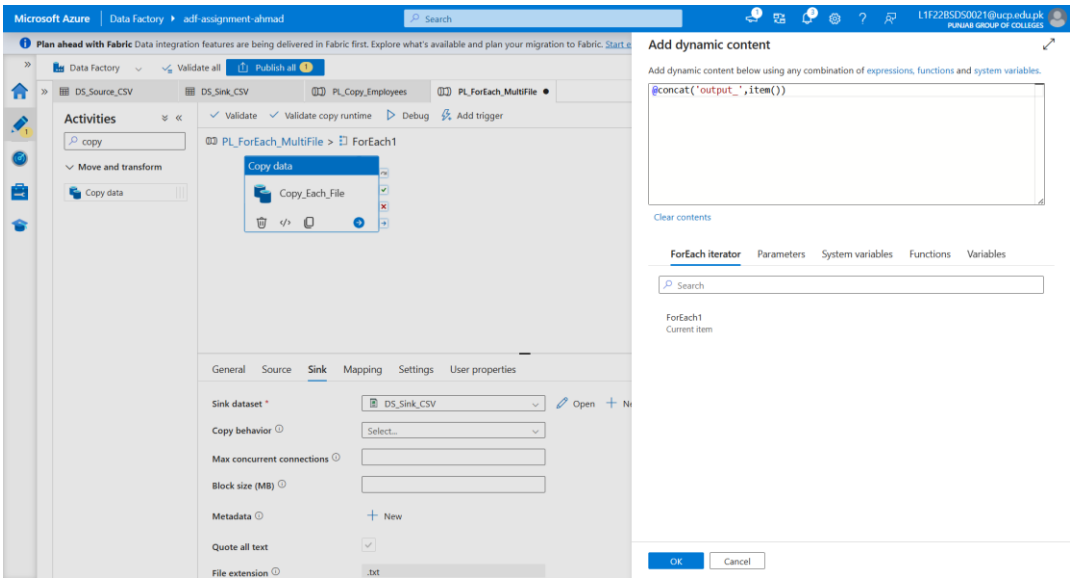
- Use:
  - If Condition
  - ForEach
  - Set Variable
  - Lookup

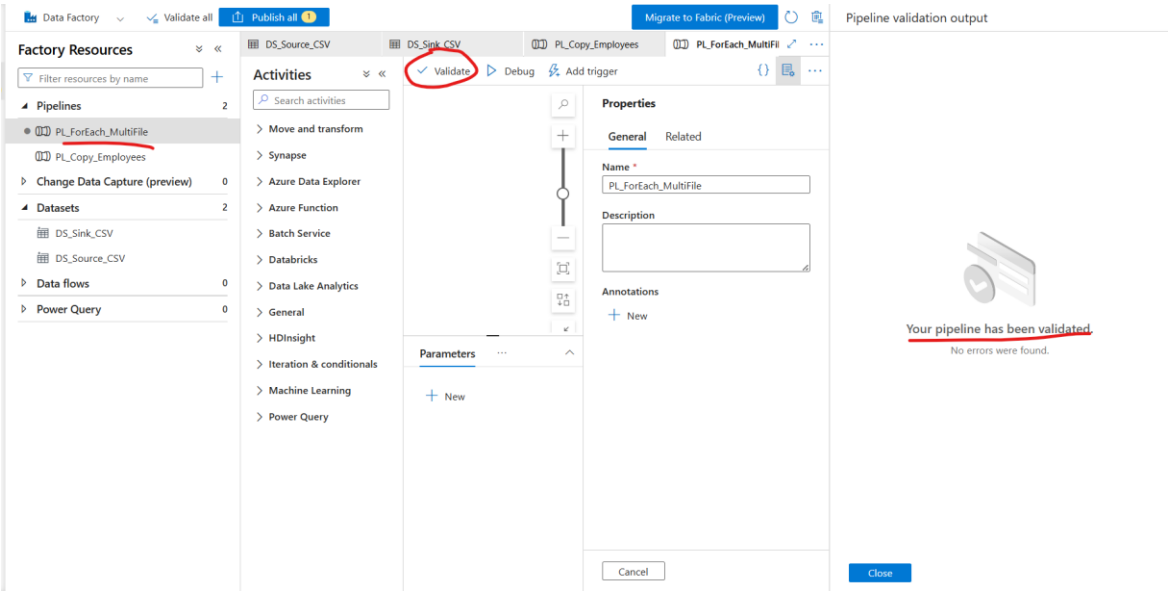
Task:

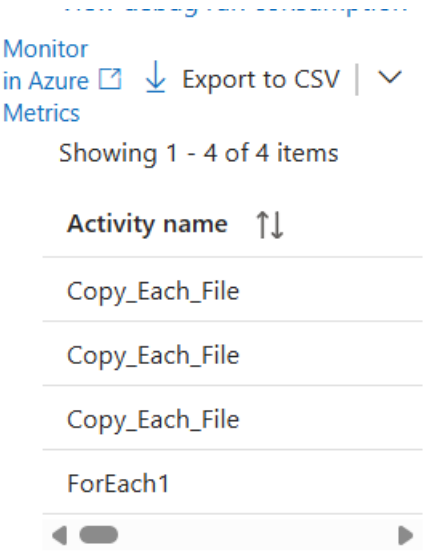
Process multiple files dynamically using ForEach loop.

Deliverables:

- Screenshot of pipeline with control flow
- Explanation:
  - Why control flow is needed
  - Business use case example





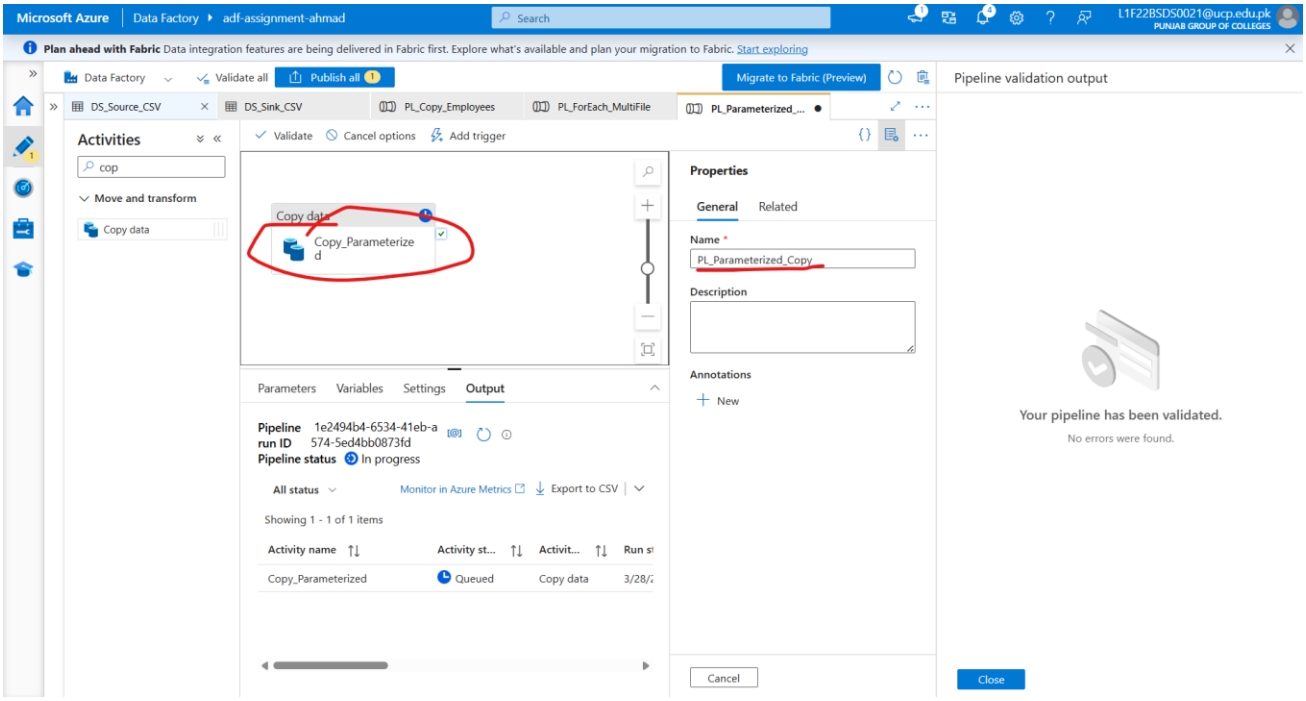


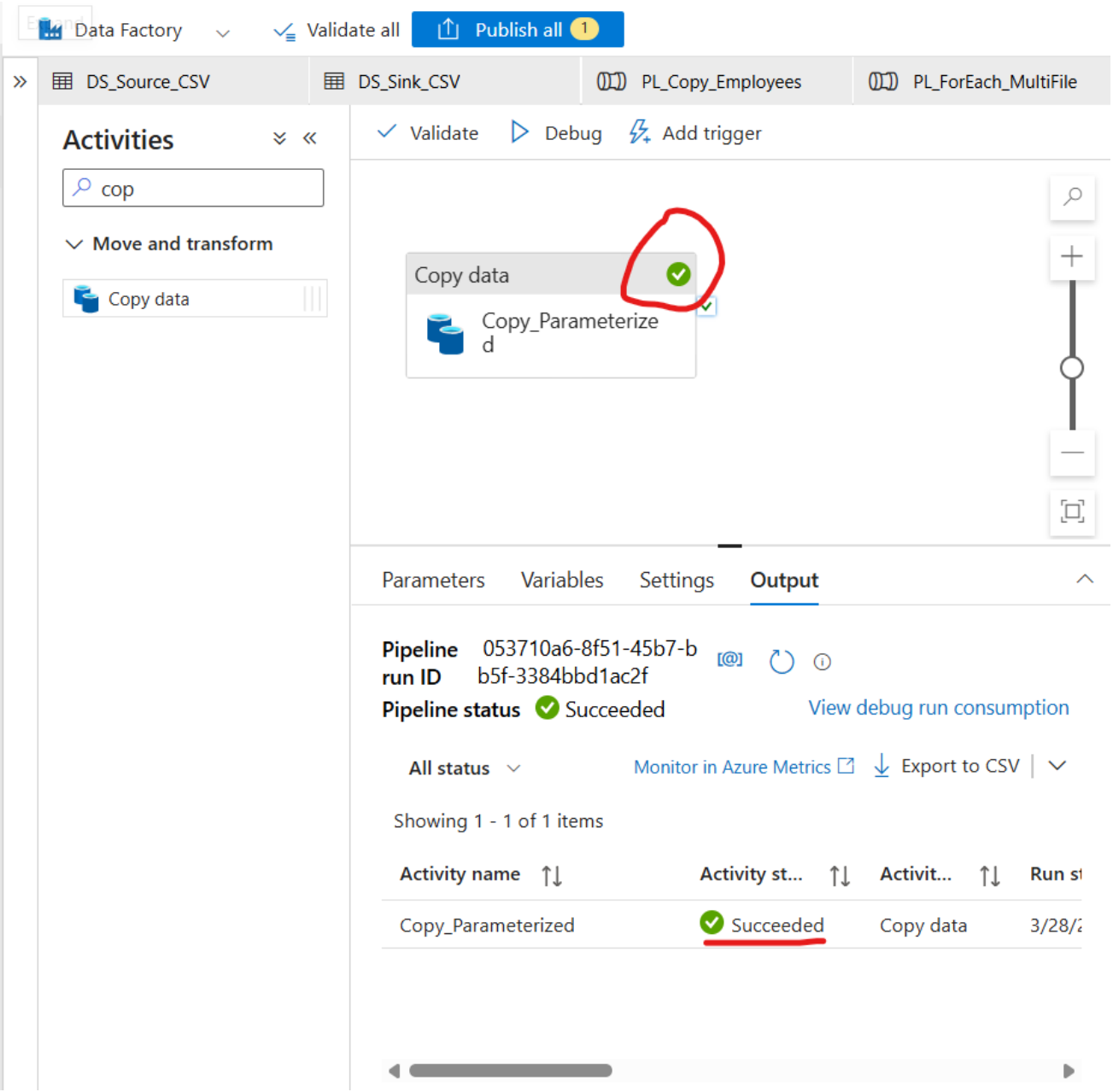
6. Parameterization & Dynamic Content

- Create parameterized pipeline
- Pass parameters dynamically

Deliverables:

- Example showing same pipeline processing multiple tables
- Documentation:
  - What is dynamic content?
  - Why static pipelines are bad practice?


Microsoft Azure
Data Factory
adf-assignment-ahmad
Search
L1F22BSDS0021@ucp.edu.pk
PUNJAB GROUP OF COLLEGES
Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. Start exploring
Data Factory
Validate all
Publish all 1
Migrate to Fabric (Preview)
Pipeline validation output
DS_Source_CSV
DS_Sink_CSV
PL_Copy_Employees
PL_ForEach_MultiFile
PL_Parameterized_...
Activities
cop
Move and transform
Copy data
Validate
Cancel options
Add trigger
Copy_Parameterized
Properties
General
Related
Name *
PL_Parameterized_Copy
Description
Annotations
New
Parameters
Variables
Settings
Output
Pipeline run ID 1e2494b4-6534-41eb-a574-5ed4bb0873fd
Pipeline status In progress
All status
Monitor in Azure Metrics
Export to CSV
Showing 1 - 1 of 1 items
Activity name
Activity st...
Activit...
Run st
Copy_Parameterized
Queued
Copy data
Cancel
Your pipeline has been validated.
No errors were found.
Close

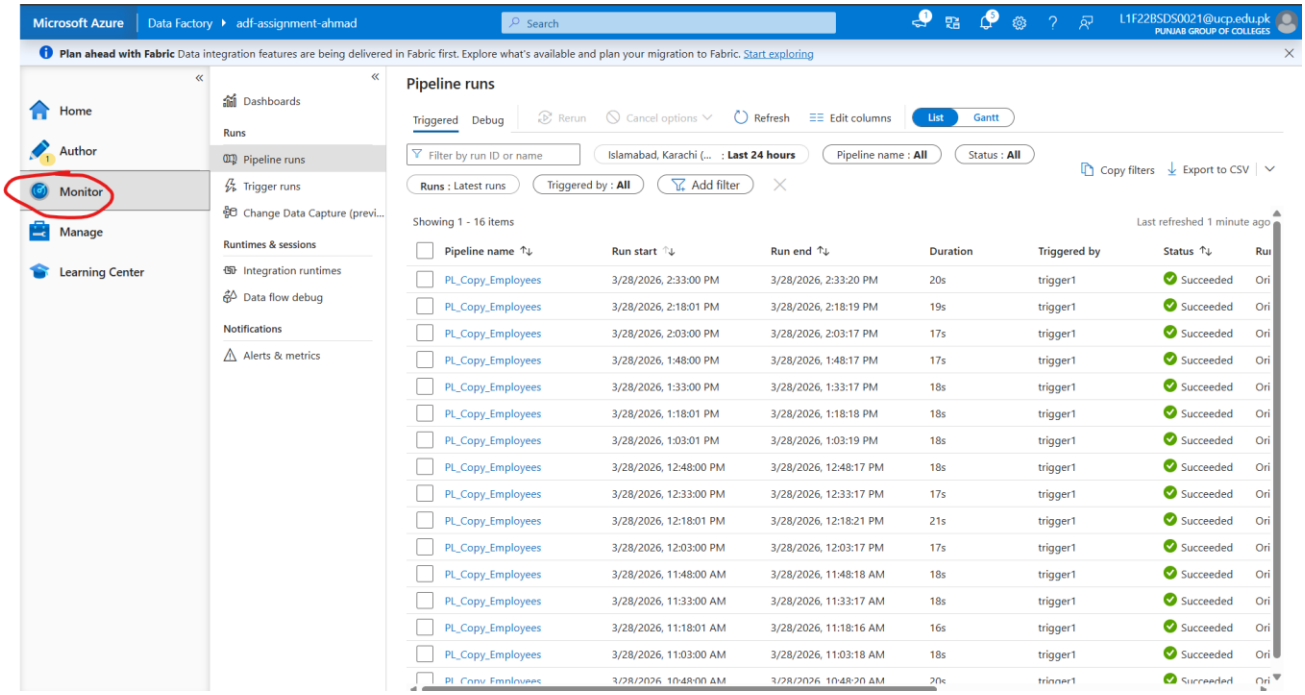


7. Monitoring & Debugging

- Use Monitor tab
- Observe success/failure logs
- Implement retry policy

Deliverables:

- Screenshot of monitor tab



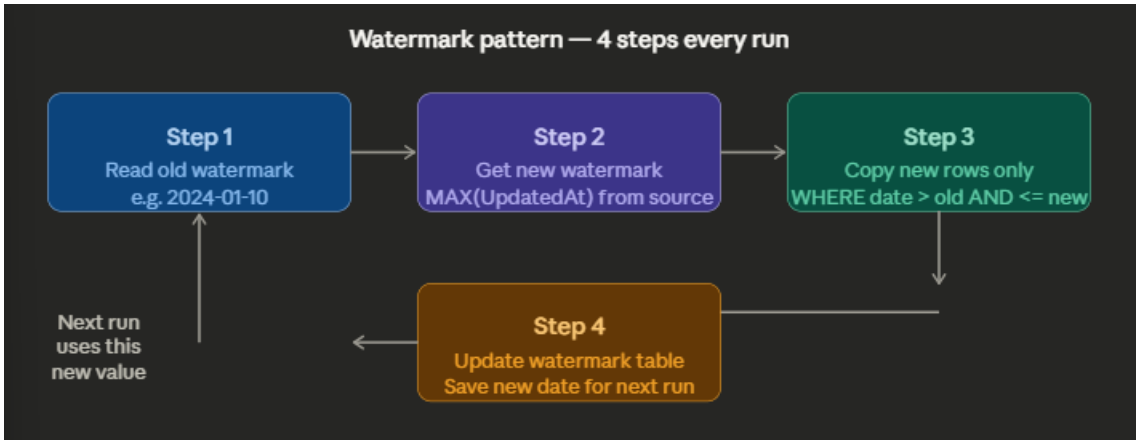
- Explanation:
  - How debugging works in ADF
  - Best practices for production monitoring

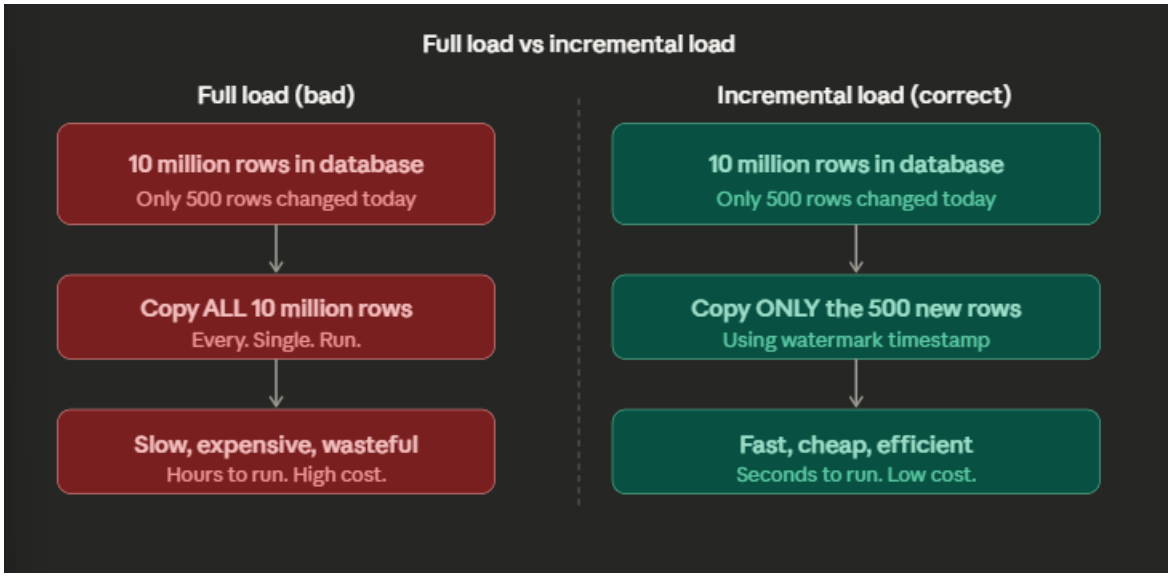
8. Incremental Load (Very Important)

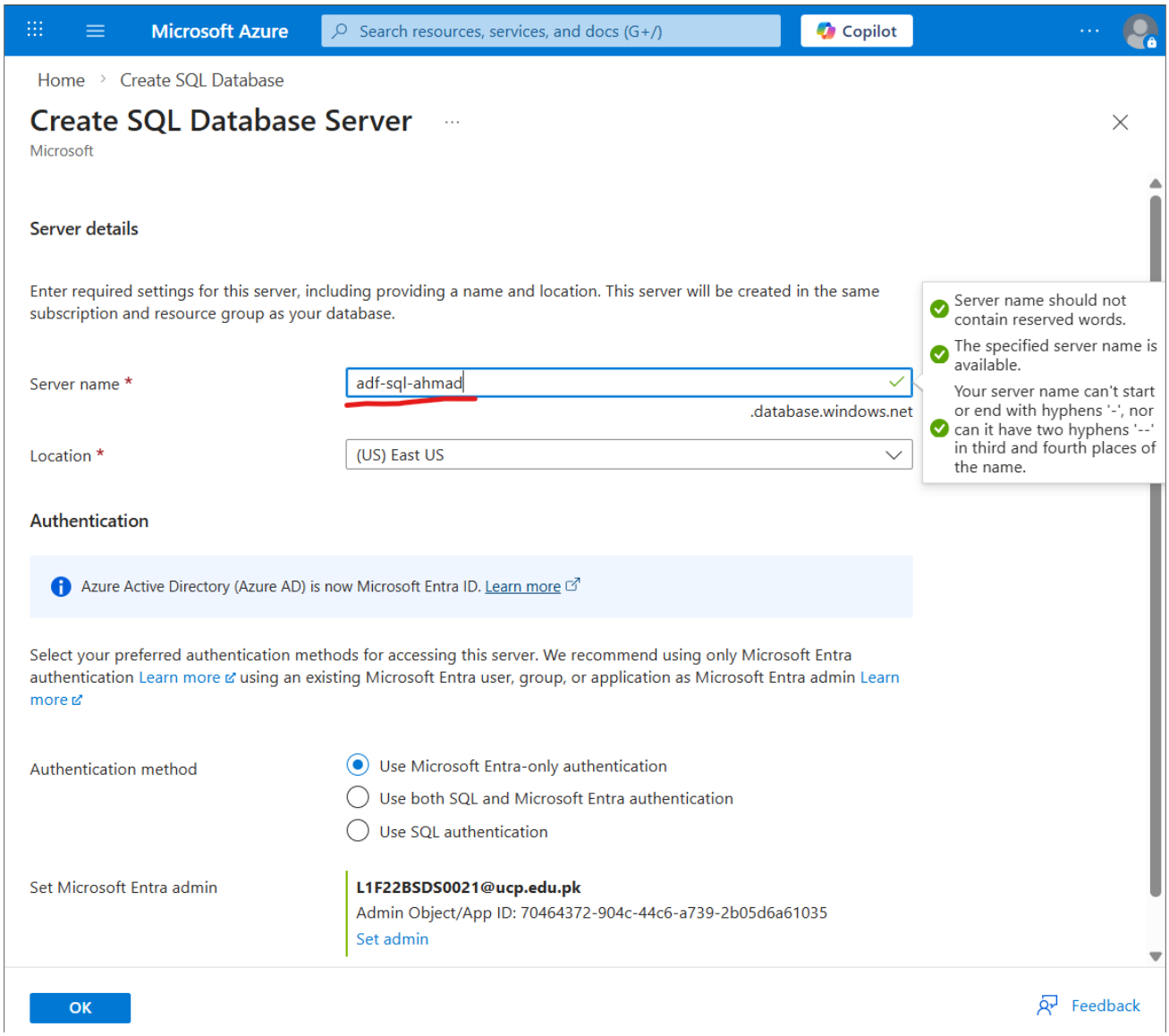
- Implement watermark-based incremental loading

Deliverables:

- Architecture explanation
- Step-by-step watermark logic
- Why incremental load is critical in real systems




Full load vs incremental load
Full load (bad)
10 million rows in database
Only 500 rows changed today
Copy ALL 10 million rows
Every. Single. Run.
Slow, expensive, wasteful
Hours to run. High cost.
Incremental load (correct)
10 million rows in database
Only 500 rows changed today
Copy ONLY the 500 new rows
Using watermark timestamp
Fast, cheap, efficient
Seconds to run. Low cost.


Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
Home
Create SQL Database
Create SQL Database Server
Microsoft
Server details
Enter required settings for this server, including providing a name and location. This server will be created in the same subscription and resource group as your database.
Server name *
adf-sql-ahmad
.database.windows.net
Location *
(US) East US
Server name should not contain reserved words.
The specified server name is available.
Your server name can't start or end with hyphens '-', nor can it have two hyphens '--' in third and fourth places of the name.
Authentication
Azure Active Directory (Azure AD) is now Microsoft Entra ID. Learn more
Select your preferred authentication methods for accessing this server. We recommend using only Microsoft Entra authentication Learn more using an existing Microsoft Entra user, group, or application as Microsoft Entra admin Learn more
Authentication method
Use Microsoft Entra-only authentication
Use both SQL and Microsoft Entra authentication
Use SQL authentication
Set Microsoft Entra admin
L1F22BSDS0021@ucp.edu.pk
Admin Object/App ID: 70464372-904c-44c6-a739-2b05d6a61035
Set admin
OK
Feedback

Create SQL Database

Microsoft

Workload environment

- ○ Development
- ◉ Production

Default settings provided for Production workloads. Configurations can be modified as needed.

Compute + storage *

Hyperscale
Standard-series (Gen5), 2 vCores, zone redundant disabled
Configure database

Backup storage redundancy

Choose how your PITR and LTR backups are replicated. Geo restore or ability to recover from regional outage is only available when geo-redundant storage is selected.

Backup storage redundancy

- ◉ Locally-redundant backup storage
- ○ Zone-redundant backup storage
- ○ Geo-redundant backup storage
- ○ Geo-Zone-redundant backup storage

Backup storage redundancy can be configured for primary database or geo replica. This configuration is not applicable for named replicas.

Review + create | Next : Networking >

Home

Microsoft.SQLDatabase.newDatabaseNewServer_c97f40ff07e44f77aad65 | Overview

Deployment

Search

- Overview
- Inputs
- Outputs
- Template

Delete | Cancel | Redeploy | Download | Refresh

Deployment is in progress

Deployment name : Microsoft.SQLDatabase.newDatabaseNewServer_c97f...
Subscription : Azure for Students
Resource group : ADF-Assignment-RG
Start time : 3/28/2026, 2:49:11 PM
Correlation ID : 2c2d956c-944b-42d6-8c68-f24becbed34d

Deployment details

Resource	Type	Status	Operation details
adf-sql-ahmad	Microsoft.Sql/servers	Accepted	Operation details

Microsoft Azure | Search resources, services, and docs (G+/) | Copilot | L1F22BSDS0021@ucp.e... PUNJAB GROUP OF COLLEGES (P...

Home

adf-sql-ahmad | Networking

SQL server

- Search
- Overview
- Activity log
- Access control (IAM)
- Tags
- Quick start
- Diagnose and solve problems
- Resource visualizer
- Settings
- Data management
- Security
  - Networking
  - Microsoft Defender for Cloud
  - Transparent data encryption
  - Identity
  - Auditing
- Intelligent performance
- Monitoring
- Automation
- Help

Add or remove favorites by pressing Ctrl+Shift+F

Selected networks

Connections from the IP addresses configured in the Firewall rules section below will have access to this database. By default, no public IP addresses are allowed. Learn more

Please save public network access value before adding new virtual networks.

Virtual networks

Allow virtual networks to connect to your resource using service endpoints. Learn more

+ Add a virtual network rule

Rule	Virtual network	Subnet	Address range	Endpoint status	Resource group	Subscription	State

Firewall rules

Allow certain public internet IP addresses to access your resource. Learn more

+ Add your client IPv4 address (39.36.68.247) + Add a firewall rule

Rule name	Start IPv4 address	End IPv4 address
ClientIPAddress_2026-3-28_19-16-14	39.36.68.247	39.36.68.247

Exceptions

☑ Allow Azure services and resources to access this server

Save | Discard

Home > ADF_DB (adf-sql-ahmad/ADF_DB)

ADF_DB (adf-sql-ahmad/ADF_DB) | Query editor (preview)

Azure SQL Database Hyperscale

- Search
- Overview
- Activity log
- Tags
- Diagnose and solve problems
- Query editor (preview)
- Mirror database in Fabric (preview)
- Resource visualizer
- Settings
- Data management
- Integrations
- Power Platform
- Security
- Intelligent performance
- Monitoring
- Automation
- Help

Add or remove favorites by pressing Ctrl+Shift+F

New query | Templates | Open in | Feedback | Sign out | Classic experience

Explorer

- Search
- ADF_DB
  - dbo
- Queries
  - SQL query 1

SQL query 1

Run | Save as view

To create table

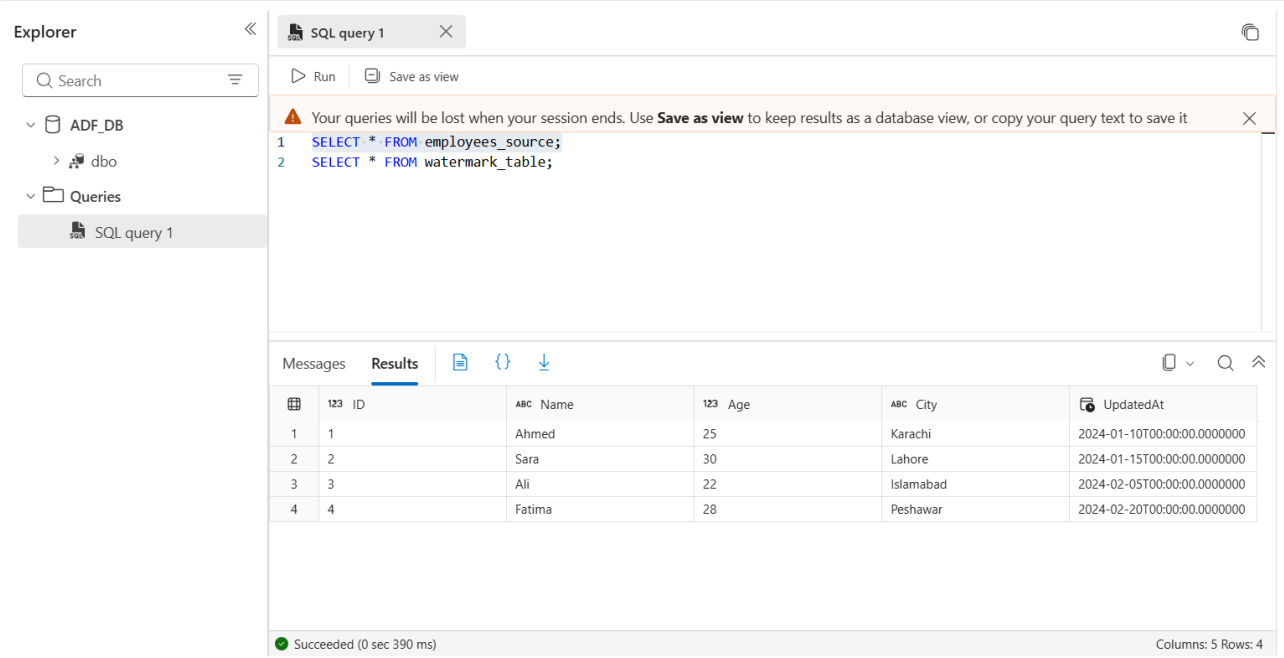
Your queries will be lost when your session ends. Use **Save as view** to keep results as a database view, or copy your query text to save it

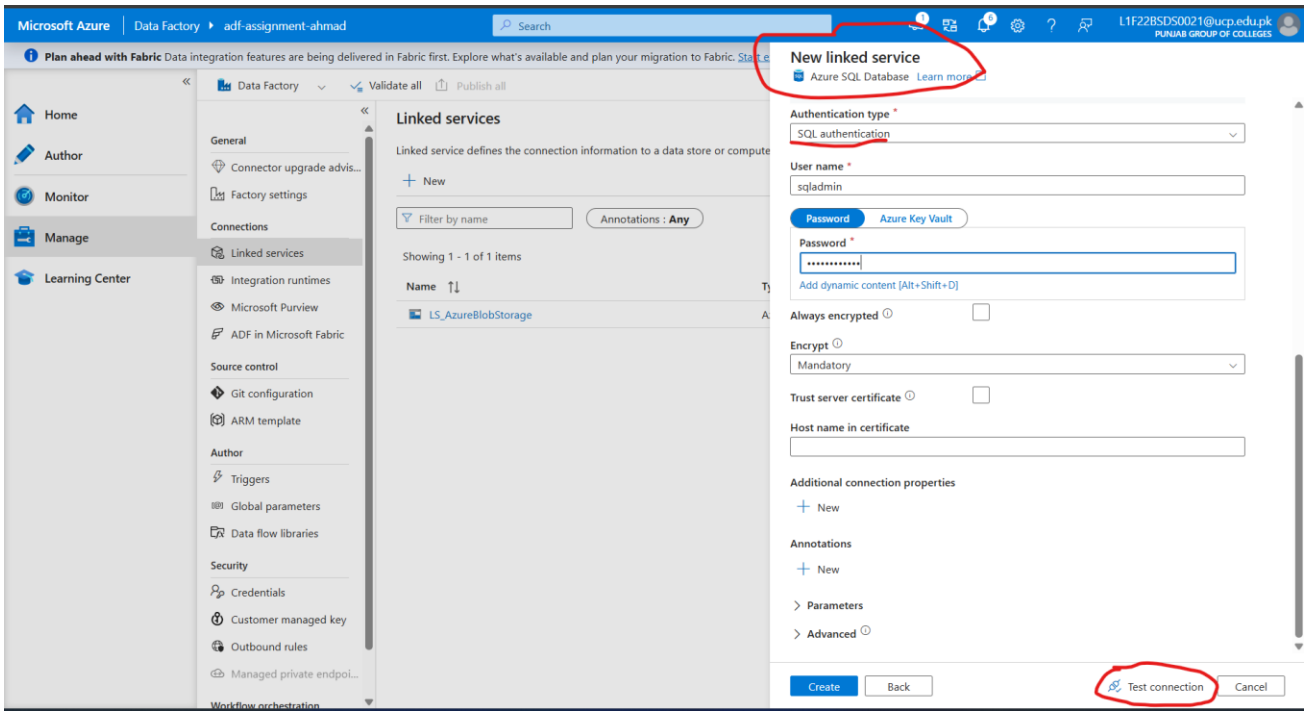
```
1 CREATE TABLE employees_source (
2     ID INT,
3     Name VARCHAR(100),
4     Age INT,
5     City VARCHAR(100),
6     UpdatedAt DATETIME DEFAULT GETDATE()
7 );
```

Messages | Copy messages

7:19:27 PM	Started executing on line 1 Query executed successfully

Succeeded (0 sec 367 ms)





Part 3 – Advanced Concepts (Based on Video 2)

Now follow:

https://www.youtube.com/watch?v=KxUm1eXQha8

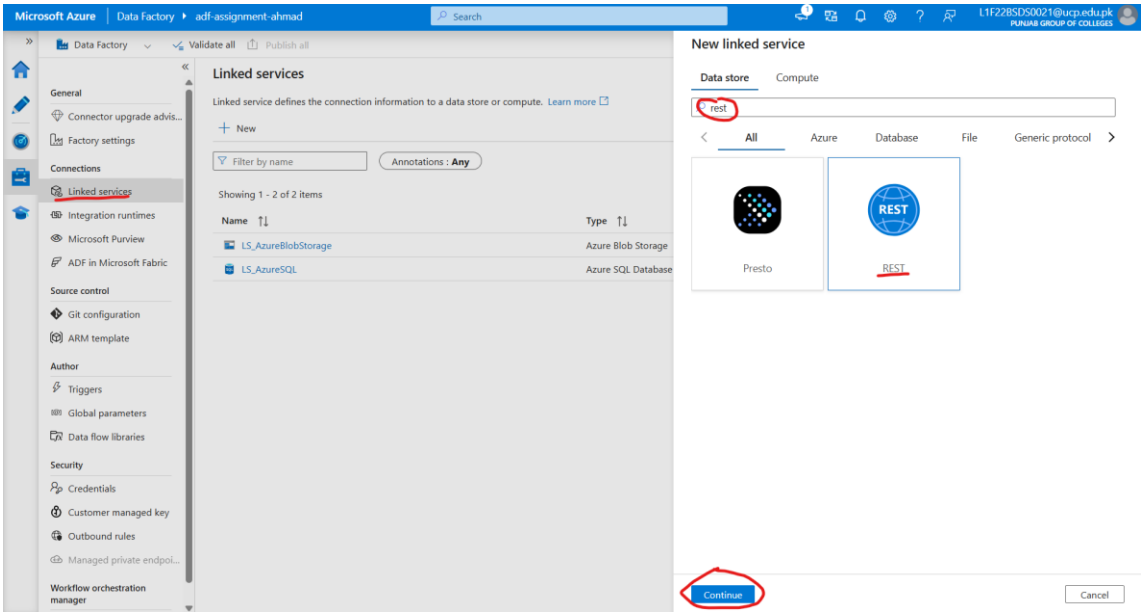
9. REST API Integration

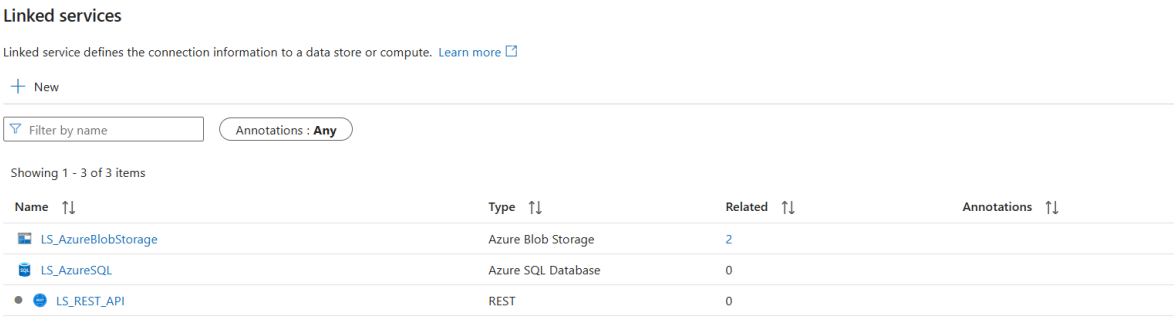
- Connect to REST API

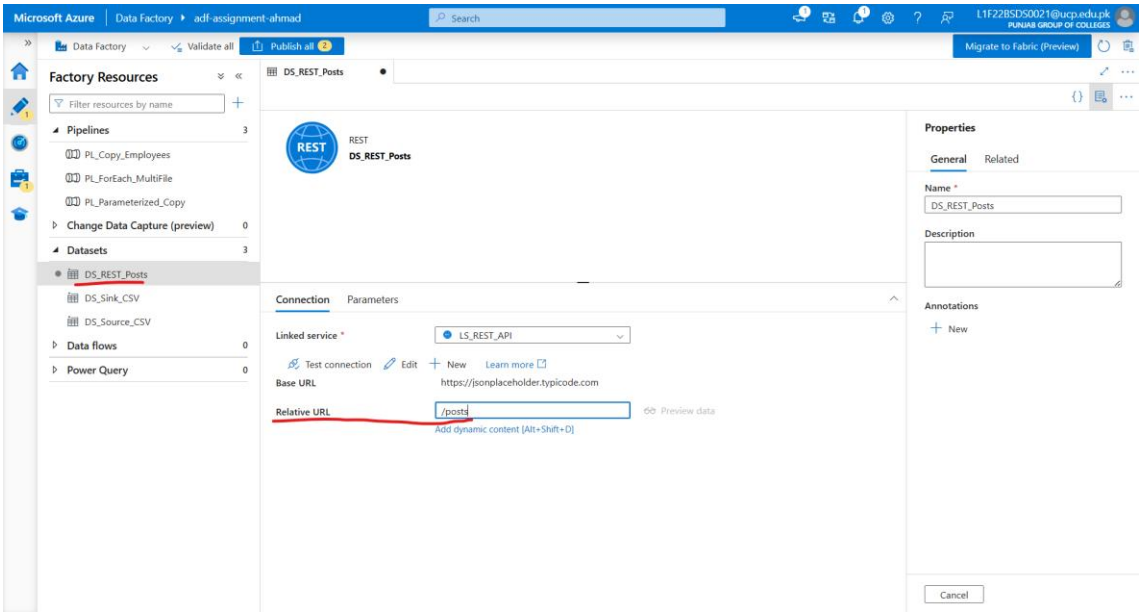
- Handle pagination (if available)
- Load into SQL

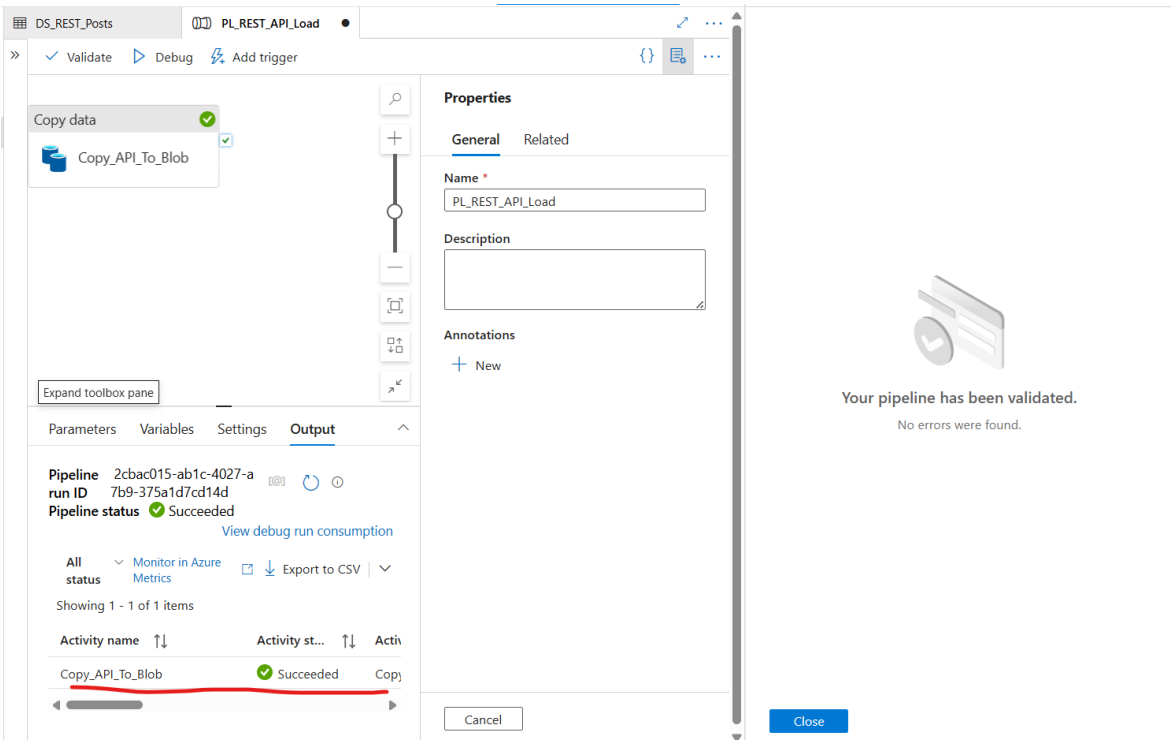
Deliverables:

- Screenshot of REST Linked Service
- Explanation:
    - Authentication
    - Pagination handling
    - Real-world scenario









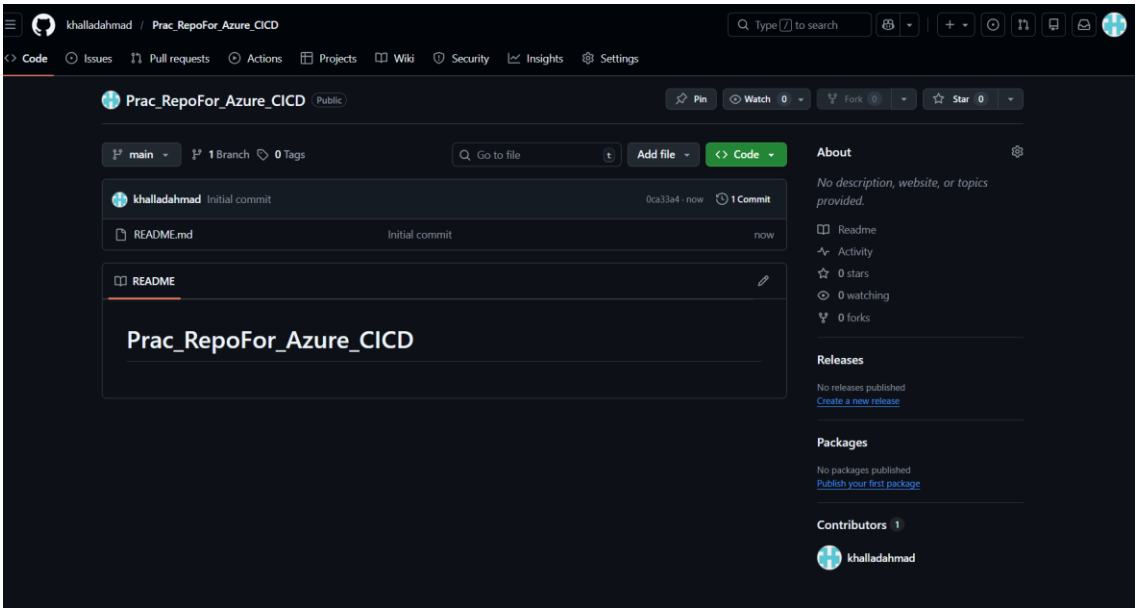
Git Integration (CI/CD)

- Connect ADF to GitHub or Azure DevOps
- Publish changes
- Create branch

Deliverables:

- Screenshot of Git integration
- Documentation:
  - Why CI/CD matters

- How version control prevents production failure



11. Advanced Looping & Error Handling

- Nested ForEach
- Error handling strategy
- Timeout and retry policy

Deliverables:

- Pipeline screenshot
- Explanation:
  - Enterprise-level pipeline design principles

12. Deployment Strategy

- Dev → Test → Production (conceptual if not possible practically)

Deliverables:

- Deployment architecture diagram
- Explanation:
  - What changes across environments?

Final Capstone Project (Compulsory)

Build an End-to-End Production-Level Data Pipeline:
Link

Scenario:

1. Ingest data from:
    - Blob Storage OR REST API
2. Transform (Control Flow + Dynamic Logic)
3. Load into Azure SQL
4. Apply incremental load
5. Add trigger
6. Add monitoring
7. Add Git integration

Mandatory Submission Requirement (Very Important)

Each student must submit:

1. Word Documentation File (.docx)

The Word file must include:

- Title page
- Table of Contents
- Architecture diagrams
- Screenshots
- Step-by-step explanation of every activity
- Error troubleshooting section
- Lessons learned
- Best practices
- Conclusion

Your documentation must be detailed enough that:

A beginner can follow your document and master Azure Data Factory WITHOUT watching any video.

If your documentation skips steps, you will lose marks.

2. ZIP Folder

Containing:

- Exported ADF ARM template (if possible)
- Screenshots
- Pipeline JSON
- Architecture diagrams

Expectations (No Surface-Level Work)

This assignment is designed to simulate:

- Real Data Engineer responsibilities
- Production pipeline development
- Enterprise documentation standards

If you complete this properly, you can confidently claim: “I can build production-grade Azure Data Factory pipelines.”